

Detecting Social Group Appeals in Parliamentary Speeches Using LLMs

Master Thesis

A thesis submitted in partial fulfilment of the requirements
for the degree of the Master of Social and Economic Data Science
at the University of Konstanz.

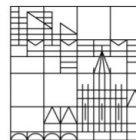
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01/1364761

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Period of completion: 13.10.2025 – 23.03.2026

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Konstanz, March 2026

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List of Abbreviations

AME	Average Marginal Effect
BES	British Election Study
BIO	Begin-Inside-Outside
BPE	Byte Pair Encoding
CoT	Chain-of-Thought
CLS	Classification Token
FP	False Positive
FN	False Negative
LLM	Large Language Model
LSD	Lexicoder Sentiment Dictionary
MI	Mutual Information
MLM	Masked Language Modeling
MP	Member of Parliament
NCE	Noise Contrastive Estimation
NER	Named Entity Recognition
NLI	Natural Language Inference
NLP	Natural Language Processing
NMI	Normalized Mutual Information
NSP	Next Sentence Prediction
SEP	Separator Token
tf-idf	term frequency-inverse document frequency
TP	True Positive
TPE	Tree-Structured Parzen Estimator
UK	United Kingdom

Chapter 1

Introduction

Representing the interests of social groups can be regarded the core of politics and is a key responsibility for political actors. Consequently, politicians regularly refer to such groups in order to mobilize them, signal responsiveness, and ultimately win their votes. Although the importance of social groups in shaping political processes and influencing electoral outcomes is widely acknowledged, the systematic study of how political actors rhetorically target them represents a relatively recent strand of political science research. Traditionally, greater attention has been devoted to the analysis of how parties seek to mobilize electoral support by promoting substantive policies they expect to resonate with their target electorate (Dolezal et al., 2014). Thau (2019) emphasizes that social group appeals, which tend to operate on a more symbolic level by generally expressing support for the group in question, are used in their own right and have been an integral part of party communication over recent decades.

The majority of studies that measure the use of social group appeals rely on manual hand-coding of party manifestos (Dolinsky, 2023; Horn et al., 2021; Howe et al., 2022; Huber, 2022). This means that the researchers need to record whether and where the party refers to a group that matches their previously established definition of a social group. Although this approach produces valid results due to human supervision, it is highly resource-intensive and therefore naturally limits the size of the research corpus. Dictionary-based approaches, in which researchers construct predefined lists of keywords that can be computationally applied to large volumes of text, represent a logical advancement. However, since appeals can be expressed in a wide variety of ways, it is difficult to anticipate and include all potentially relevant keywords in such a dictionary. Large Language Models (LLMs) have been shown to be much better suited to tasks that require rich language understanding and have already been successfully applied to the detection of social group appeals (Horne et al., 2024; Licht & Szczepanski, 2025). These applications achieve generally satisfactory results, but can be improved in a variety of ways. None of them comparatively evaluates which LLM-type might be best suited for the task. Furthermore, they either classify only the general presence of a social group appeal without encoding the author’s positional stance, or they rely on a strictly fixed set of social groups that an appeal might target.

In this paper, I use a narrow definition of what constitutes a social group and, following Huber and Dolinsky (2023), conceptualize it as a segment of society characterized by shared or acquired sociodemographic characteristics. A social group appeal is defined as an intentional act that associates a political actor with a social group in a supportive, negative, or neutral manner. I systematically evaluate and compare the effectiveness of both encoder-only and generative LLMs for the detection of social group appeals. I develop separate LLM-based classification models that identify the exact span of a mention within political texts and classify the author’s positional stance toward the referenced group. In addition, I propose to apply k-means clustering to the vector representations of the detected mentions to inductively categorize the appeals into qualitative social group categories. This unsupervised clustering approach has the distinct advantage of revealing social group clusters that might not have been anticipated before and

therefore could not be included in a supervised classifier. To further improve the separability of the input representations, I employ contrastive learning techniques which leverage previously known social group categories and example phrases collected in an expert dictionary. This procedure not only substantially improves the cluster structure, but also ensures to recover the known group categories while still allowing for new ones to show up.

I demonstrate the usefulness of this methodological approach by applying it to parliamentary speeches in the United Kingdom’s (UK) House of Commons to examine the use and effects of strategic positive appeals to different age groups. British politics was structured around economic class as a primary organizing force for party competition and voting behavior for much of the previous century. However, beginning with Labour’s reorientation toward middle-class voters in the 1990s, these traditional ties have loosened substantially, and economic groups are now addressed only rarely in British parties’ communication (Evans & Tilley, 2017). With the declining importance of economic status, age has emerged as one of the strongest predictors of voting behavior in the UK (Serra, 2024). Younger voters predominantly support the Labour Party, whereas older voters mostly favor the Conservatives. This pattern, already termed the “age cleavage” (Curtice (2023)), is most often attributed to ideological differences between the age groups, which have become particularly salient in recent years (Serra, 2024). Consequently, these contrasting opinions, for instance on cultural issues, exert an increasing influence on people’s voting behavior.

In this work, I empirically examine the extent to which this age divide is reflected in the major British parties’ parliamentary rhetoric. I further analyze the effect of positive age-based appeals on British peoples’ voting behavior. Using the developed classification pipeline, I identify all appeals directed at young and elderly voters within parliamentary question debates in the UK House of Commons between 2010 and 2019. I expect Labour legislators to appeal more frequently to young people, whereas Conservative members of parliament (MPs) are assumed to focus more on elderly voters. Furthermore, I anticipate that legislators from both parties will be influenced by the demographic age composition of their local constituencies in their use of age-based appeals. Hence, Labour MPs should increase positive appeals toward young voters in constituencies with a larger proportion of young residents, while Conservative MPs should do the same for elderly voters, reflecting a strategic adaptation to the age profile of their electorates. I test the effect of age-based appeals on voting behavior using survey data from the British Election Study (BES). I link each MP’s proportion of age-based appeals to survey respondents in their constituency and analyze whether these appeals influence the age gap in party support between young and old voters. I expect that existing baseline differences in party support are reinforced by strategically employed positive appeals toward age groups.

The comparison of different model families for detecting social group appeals shows that encoder-based BERT models generally outperform generative LLMs in identifying both the presence and the span of a group reference, as well as the speaker’s stance toward the group. Stance classification benefits considerably of using a BERT model that has already been fine-tuned on a diverse range of Natural Language Inference (NLI) tasks. Dictionary-based methods are only somewhat competitive in detecting the general presence of an appeal but perform substantially worse in identifying the exact sequence span and the positional stance. The application of k-means clustering to classify group appeals into qualitative categories yields promising results, as it allows for the recovery of known social group categories while also enabling the detection of previously unknown ones. The results further indicate that this approach benefits substantially from the prior application of contrastive learning to the embedding model, without which the clustering produces less distinct and less meaningful groupings.

The analysis of age-based rhetoric in the UK House of Commons indicates that such appeals are used less extensively by political parties than expected. Both Labour and Conservative MPs exhibit similar baseline rates of appealing to young people, while Labour MPs address elderly voters even more frequently than their Conservative counterparts. Moreover, I find that parties

are more responsive to increases in the share of an age group within a constituency if that group lies outside their core electorate. This pattern may reflect the greater electoral gains to be made among voters who are typically less aligned with the party. Finally, the results suggest that age-based appeals have some potential to influence the age divide in party support. While youth-oriented appeals by Labour legislators increase young voters' probability of supporting Labour, appeals to elderly voters by Conservative legislators appear to discourage young voters from supporting the Conservative Party. However, most estimated effects are accompanied by considerable uncertainty and should therefore be interpreted with caution.

The thesis is structured as follows. In Chapter 2 I outline the methodological background of continuous text representations, LLM architectures, contrastive learning and clustering algorithms. In Chapter 3 I review the relevant related work which encompasses the substantive concept of social group appeals, existing methods for detecting them as well as the role of parliamentary speechmaking in politicians' attempts to gain electoral support. Chapters 4 and 5 present the methodology and evaluation results of the classification and clustering models. In Chapters 6 and 7 I apply the developed method to parliamentary question debates in the British parliament and analyze the usage patterns and effects of age-based appeals. Chapter 8 discusses both the methodological and empirical results and Chapter 9 concludes the thesis by summarizing the main findings, addressing limitations and identifying potential avenues for future research.

Chapter 2

Background

Recent advancements in Natural Language Processing (NLP) driven by the revolutionizing success of LLMs based on the Transformer architecture provide promising possibilities to detect social group appeals in political text (Horne et al., 2024; Licht & Szczepanski, 2025). In this chapter I give an overview on the core functioning of these models as well as the conceptual differences between them. Furthermore, I outline the concepts of contrastive learning and the application of clustering algorithms to text vector representations. This methodological background is essential for understanding the development and evaluation of the classification and clustering models used in this thesis.

2.1 Text Representations and Large Language Models

In the following, I outline how LLMs internally represent and process text. In addition, I illustrate the key distinctions between different LLM architectures and describe how they are used differently for tackling classification tasks.

2.1.1 Subword Tokenization

LLMs process text not as whole words, but as subword tokens from which the original words can be reconstructed. These tokens, stored in a fixed vocabulary, may correspond to original words or to smaller units that can be combined to form many different words. These models rely on such tokenization for several reasons. The dynamic composition of words by using many subword tokens makes it unnecessary to store all unique words the language model might possibly encounter in its vocabulary. The size of this vocabulary would otherwise be immensely large. The LLM reduces the chance of words outside of the the vocabulary, by working with subpieces from which it is possible to construct the majority of occurring words. Subword tokenization represents a compromise between working with a theoretically infinite vocabulary containing all possible words existing and only individual characters. It would also be possible to create every imaginable word from individual characters only, but this would require more computational effort and neglect semantic structure that is encoded into the words themselves.

The vocabulary of subword tokens gets produced by the application of a tokenization algorithm on a training corpus. Prominent examples of these algorithms are WordPiece (Wu et al., 2016) and Byte Pair Encoding (BPE) (Sennrich et al., 2016) both of which are very similar in terms of their training process. First, the training corpus gets preprocessed by splitting words according to whitespaces and other predefined rules. Then, all words get split up into individual characters and added to the vocabulary alongside frequency counts capturing how often they occur in the corpus. The algorithm considers all potential merges of these individual characters and assigns each merge a score based on how frequently these characters occur as a pair in the corpus, normalized by the product of the overall occurrence of the characters forming the merge.

The pair with the highest score is added to the vocabulary and the counts get updated. This process continues until the desired vocabulary size is reached. The result of this process is a vocabulary which captures the most common forms of character combinations as they occur in the training corpus. During inference, input tokens are decomposed into subword units using the fixed vocabulary. WordPiece performs greedy longest-match segmentation over the vocabulary, whereas BPE starts from a character-level representation and iteratively applies the learned merge rules until no further merges are possible.

2.1.2 Embedding Representations

Whereas natural language is inherently discrete, LLMs map words to continuous numerical vector representations that encode semantic information. These word vectors are learned such that words with similar usage patterns are located close to one another in a high-dimensional vector space. The standard way of assessing the similarity between two word vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ is by calculating the cosine similarity between them.

$$\cos(\theta) = \frac{\mathbf{v}_1 \cdot \mathbf{v}_2}{\|\mathbf{v}_1\| \|\mathbf{v}_2\|} \quad (2.1)$$

The numerator is the dot product of the two vectors and the denominator is the product of their Euclidean lengths. This normalization constrains the two vectors to unit length, making cosine similarity sensitive to the vectors' relative distributions instead of their absolute magnitudes. The similarity measure corresponds to the cosine of the angle θ between the two vectors (Manning et al., 2008). An alternative way of measuring the similarity between two vectors is by computing the distance between them. A commonly used distance measure is the Euclidean distance, which for two vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ is defined as:

$$d(\mathbf{v}_1, \mathbf{v}_2) = \|\mathbf{v}_1 - \mathbf{v}_2\|_2 = \sqrt{\sum_{i=1}^n (v_{1,i} - v_{2,i})^2} \quad (2.2)$$

Analogous to cosine similarity, both vectors can be normalized to unit length before calculating the distance in order to make the comparison independent of their absolute magnitudes.

The conceptual idea for word embeddings is rooted in the distributional hypothesis which states that the meaning of a word can be inferred from its direct surrounding context (Firth, 1957; Harris, 1954). Early approaches utilize this insight by creating word vectors based on word entries in a word-context co-occurrence matrix. Words that are used in similar contexts consequently have similar row entries in this matrix. The problem of this approach is the sparsity of the matrix as most words have a quite narrow set of potential context words so that the vast majority of entries is zero (Turney & Pantel, 2010). A promising approach for representing an entire document as a numeric vector is to compute the term frequency-inverse document frequency (tf-idf) score for each term appearing in the document. The tf-idf value for term t in document d is defined as:

$$\text{tf-idf}_{t,d} = \text{tf}_{t,d} \times \text{idf}_t \quad (2.3)$$

Here, tf denotes the term frequency which is the number of times term t occurs in document d , while the inverse document frequency is given by:

$$\text{idf}_t = \log \frac{N}{\text{df}_t} \quad (2.4)$$

N is the total number of documents in the corpus and df_t is the number of documents in which term t appears. Higher tf-idf values indicate greater relevance of a term to a particular document. A document can thus be represented as a vector of tf-idf scores over all terms in the corpus.

Bengio et al. (2003) influentially introduces the idea of learning dense word vector representations as part of a neural language model. Their model is trained to predict the next word in a sequence based on a fixed number of preceding words, which are represented using one-hot encodings and projected into a continuous embedding space via a shared weight matrix. Through training, the model learns vector representations for words that capture regularities in their usage. This architecture enables generalization beyond observed word sequences, as it can assign similar representations to words occurring in similar contexts, allowing the model to make plausible predictions for sequences not encountered during training.

Building on the same idea of encoding a word’s contextual usage patterns into a fixed-dimensional vector representation, Mikolov, Chen, et al. (2013) and Mikolov, Sutskever, et al. (2013) propose a simplified neural model for learning word embeddings. Similar to Bengio et al. (2003), their word2vec model predicts a target word from its surrounding context. They simplify the model architecture by removing the non-linear hidden layer to efficiently train the model on large text corpora. As a result, they observe that the learned embedding space captures both syntactic as well as semantic regularities. Relationships between words can be expressed as linear transformations in the vector space. Pennington et al. (2014) later leverage the global word co-occurrence matrix to compute word embeddings with the same properties without using local context windows.

These word embeddings generally capture a word’s semantics well, but they are inherently static. Each word is represented by a single vector that aggregates its usage across the entire corpus, making such embeddings unable to account for variations in meaning that arise in different contexts. More recent approaches address this limitation by producing contextualized word embeddings (Devlin et al., 2019; Peters et al., 2018; Radford et al., 2018). In these models, word representations are dynamically computed based on the surrounding words in a sentence, allowing tokens to influence one another through contextual interactions. As a result, the final word vectors are dependent on their surrounding context and significantly better suited for downstream tasks. The core mechanisms underlying these models are explained in the following sections.

2.1.3 Encoder Models

All modern LLMs are based on the Transformer model (Vaswani et al., 2017). This is a deep neural network that uses the self-attention mechanism to compute token dependencies in a text sequence. It does that very efficiently through parallel computing which allows the model to process large amounts of text data in the training phase. The self-attention mechanism produces contextualized word representations as each word can attend to other words in the sequence regardless of the positional distance between them. Transformer-based language models can be distinguished by the kind of attention mechanism used as well as the model’s training objective. First, I outline the main principles of encoder-only models such as BERT and variants of it (Devlin et al., 2019; He et al., 2021; Liu et al., 2019; Sanh et al., 2020). These models use only the encoder-part of the original Transformer model and are usually trained on two different objective functions. For the Masked Language Modeling (MLM) objective, a certain proportion of input tokens are replaced by a special mask ([MASK]) token. The model feeds the entire sequence through the Transformer stack as well as a feedforward neural network and tries to guess the original token at the position of the [MASK] token. Within the self-attention mechanism, each token can attend to all other tokens in the input sequence. This bidirectionality allows the model to encode a semantically rich representation of the input sequence, but prevents it from actually generating text, as each token can attend to both past and future tokens. The contextualized embedding of the [MASK] token is then transformed by a learned linear projection layer with output dimensionality equal to the size of the vocabulary to produce a vector of unnormalized scores. Applying the softmax function converts these scores into a probability distribution over the vocabulary, from which the original token is predicted.

Additionally, BERT is trained using the Next Sentence Prediction (NSP) objective. In this task, the model is presented with pairs of sentences separated by a special separator ([SEP]) token. The objective is to predict whether the second sentence is the actual continuation of the first sentence as they occur in the training corpus. Each input sequence is prepended with a classification ([CLS]) token and fed into the model. Similar to the MLM objective, the contextualized representation of the [CLS] token is passed through a linear layer to produce unnormalized scores, which are then converted into a probability distribution using the softmax function. The goal of this training objective is to learn a vector representation that encodes the semantic of the entire input sequence.

Both training objectives enable BERT to acquire rich language understanding capabilities. They are commonly regarded as self-supervised learning objectives, since all supervision signals are derived directly from the training data without requiring any human annotation. This allows the model to be trained on large-scale text corpora. Building on advances in transfer learning, this training procedure serves only as a pretraining phase for the LLM. To leverage BERT’s deep bidirectional text representations, the pretrained model can subsequently be fine-tuned by adding a lightweight, task-specific output layer. Depending on the downstream task, this output layer operates at the token or at the sequence level using the [CLS] representation. This allows for token-level applications such as part-of-speech tagging or classification tasks that make a prediction about the entire text sequence as in sentiment analysis. This approach makes BERT broadly accessible, as it enables users to adapt the model to a wide range of tasks with relatively modest computational resources.

Over time, a number of variants of the BERT model got released that improve on the original by changing several aspects. Liu et al. (2019) show that BERT has been significantly undertrained and can be improved by training the model longer, with more data and bigger batches. Moreover, they remove the NSP objective as they find that it has only a marginal impact on model performance. They release their improved model under the name RoBERTa. DistilBERT (Sanh et al., 2020) is a lightweight variant of the BERT model that builds on the idea of model distillation (Hinton et al., 2015). Here, a comparatively small student model learns to mimic the behavior of a larger teacher model and is consequently able to perform on a similar level with less parameters. He et al. (2021) improve upon previous BERT variants by introducing a disentangled attention mechanism. Unlike the original Transformer architecture, DeBERTa does not encode absolute positional information directly into token embeddings. Instead, each token is represented by two vectors that encode its content and its position separately from one another. The self-attention mechanism operates on these disentangled representations by computing attention scores based on both the content of the words and their relative positions.

2.1.4 Generative Models

The second class of Transformer-based LLMs comprises generative models. These models either consist solely of the decoder component of the Transformer (Brown et al., 2020; Radford et al., 2018, 2019), or of both an encoder and a decoder (Raffel et al., 2023). In contrast to encoder-only models, generative LLMs are capable of producing text by being trained with the objective of predicting the corresponding next token given the previous tokens in a text sequence. To enable autoregressive text generation, they employ a causal self-attention mechanism in which each token can attend only to preceding tokens and not to future ones. This constraint is enforced by applying a causal mask to the attention matrix, which assigns a value of $-\infty$ to the attention scores corresponding to future tokens. To signalize when a sequence starts and should end, the model works with special tokens that mark the start ([SOS]) and end of the sequence ([EOS]). The model is trained to maximize the conditional probabilities for the next token as it occurs in the training corpus. The conditional probability of the next token is computed by applying the softmax function on the contextualized vector representation of the last token in the sequence. After training, the model is able to generate text based on a user-provided

prompt. To decide which token is appended to the growing generated sequence, the model may either select the token with the highest conditional probability or sample a token from the predicted probability distribution (Jurafsky & Martin, 2026). Users can further control the generation process by applying temperature scaling, in which all logits of the unnormalized scores are divided by a temperature parameter before applying the softmax function. Lower temperature values produce more peaked distributions and more deterministic outputs, while higher values increase randomness and diversity in the generated text.

Radford et al. (2019) demonstrate that generative LLMs of a sufficient size obtain generalization capabilities that make them perform well on a diverse downstream tasks without being explicitly fine-tuned on them. By applying zero-shot learning, they let the model predict a text answer to a task outlined in the input prompt. The model does not receive any examples on how to solve the task, but is able to do that only based on the internalized knowledge acquired by being confronted with large amounts of text data during pretraining. Brown et al. (2020) build up on these findings and show that model performance improves even further when the model gets prompted in a few-shot setting during which the model sees a few examples on how to solve the task in its input context. Still, this kind of learning procedure does not require any real training dataset, but only some positive examples on how to solve the task.

The pretraining objective of maximizing the conditional probability of the next token enables models to continue existing text sequences but does not equip them to respond effectively to custom user prompts. To address this limitation, generative LLMs undergo a second pretraining step, called instruction tuning, during which they are trained to generate text conditioned on prompts framed as instructions (Chung et al., 2024; Wei et al., 2022). During this process, the model is fine-tuned to perform well on a diverse set of tasks expressed in instructional form. As a result, the model becomes significantly more interactive across different prompt formulations while retaining the ability to generalize across a wide range of task settings. Further improvements can be achieved through careful prompt design, most notably via Chain-of-Thought (CoT) prompting (Wei et al., 2023). In this approach, the model is not only shown examples of correct answers in its context, but also demonstrations of the intermediate reasoning steps leading to those answers. Accordingly, the model first generates an explicit reasoning chain and subsequently derives the final answer from it. This prompting strategy substantially improves performance on complex tasks, but is effective only for models of sufficient scale.

2.2 NLP Task Formulation

The detection of social group appeals can be divided into two distinct tasks. The first is to identify the sequence span in which a social group is mentioned, while the second concerns classifying the author’s stance toward the referenced group. These tasks correspond to the NLP tasks of sequence labeling and stance classification which I outline in the following.

2.2.1 Sequence Labeling

Sequence labeling consists of assigning each word in a sequence an output label so that the output sequence of tags has the same length as the input sequence of words (Jurafsky & Martin, 2026). Importantly, the assignment of an output label does not depend solely on the respective word, but also on the global context of the entire sequence. Classical methods such as Hidden Markov Models (Rabiner, 2002) and Conditional Random Fields (Sutton & McCallum, 2012) explicitly model these global dependencies as a Markov chain by defining transition probabilities between neighboring labels and emission probabilities conditioned on the observed tokens. Due to the self-attention mechanism in which token vector representations get updated with respect to the other token representations, Transformer-based LLMs take these global dependencies into account as part of their architectural design. With different task designs, both encoder-only as

well as generative LLMs are consequently able to predict output tags for all individual words.

A prominent application of sequence labeling is Named Entity Recognition (NER), in which word-level tags are assigned to mark spans of text that contain mentions of entities with proper names. Hence, not all words in the sequence are necessarily of interest, but usually still all words get tagged to indicate where a named entity reference starts and where it ends. The goal of classifying social group mentions in text is close to the task of NER. Unlike NER, social groups are not necessarily named entities, but can also take more general types. I define the concept of a social group in more detail in Chapter 4. Nonetheless, social group mention detection mimics NER as the classification model needs to clearly mark text spans of entity references.

To classify text spans as entities while assigning each token an individual output tag, one needs to set up a tagging scheme that is able to unambiguously determine where a span starts and where it ends. The most prominent tagging scheme to achieve that goal is Begin-Inside-Outside (BIO) tagging (Ramshaw & Marcus, 1995; Tjong Kim Sang & De Meulder, 2003). Here, each token gets classified according to being the begin (“B”) of an entity mention, inside one (“I”), or falling outside of any mention (“O”). If an entity mention is classified, the entity category gets appended to the “B” and “I” tags. Based on these tags, it is possible to clearly separate entity spans from one another even if they are located directly next to each other in the text sequence.

To evaluate the performance of sequence labeling models that predict entity spans, standard machine learning metrics such as precision, recall, and F_1 -score are commonly applied (Jurafsky & Martin, 2026). The unit of evaluation is typically the entity rather than the individual token. Consequently, a prediction is considered correct only if the predicted span exactly matches the span of a ground-truth entity (Ramshaw & Marcus, 1995). Such a prediction is counted as a True Positive (TP), while a predicted span that does not match any ground-truth entity is considered a False Positive (FP). Conversely, a ground-truth entity span that is not detected by any prediction constitutes a False Negative (FN). For each entity class independently, precision is defined as the proportion of TP predictions among all predicted spans, while recall measures the proportion of ground-truth entity spans that are correctly identified by the model. The F_1 -score is computed as the harmonic mean of precision and recall.

$$\text{Precision} = \frac{|\text{TP}|}{|\text{TP}| + |\text{FP}|} \quad (2.5)$$

$$\text{Recall} = \frac{|\text{TP}|}{|\text{TP}| + |\text{FN}|} \quad (2.6)$$

$$F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (2.7)$$

These evaluation metrics can be regarded as a strict measure of model performance because the entire predicted span needs to match the ground truth data. Partial matches are not rewarded in any way as a prediction that only partially matches the correct span counts both as a FP prediction in the computation for the precision as well as a FN in the computation for the recall score.

2.2.2 Stance Classification

Sequence labeling addresses the detection of social group references within a text sequence, but does not provide information about the positional tone the author adopts toward these groups. The classification of this position is necessary for capturing the entire concept of a social group appeal. The positional tone expressed in text is commonly analyzed using sentiment analysis. However, this approach is not suitable for this case, as sentiment analysis focuses on the overall sentiment of a text document, whereas the objective here is to capture the specific stance

taken toward a mentioned social group. The latter is more appropriately addressed through stance classification. Particularly in the social science literature, these two approaches are often confused with one another, with sentiment analysis being applied even when the actual research focus is about the stance toward a specific entity (Bestvater & Monroe, 2023). While sentiment analysis assigns a single sentiment label to an entire text, stance classification aims to assign a stance label to each entity of interest (Küçük & Can, 2020). Thus, sentiment analysis captures whether an entire text document is written in a negative, neutral, or positive tone, whereas stance classification determines whether the speaker is in favor of a target entity, against it, or expresses neither position.

Stance classification is regarded as a supervised classification problem and has been traditionally tackled by careful feature engineering to train supervised machine learning classification models such as support vector machines or logistic regression models (Nazir et al., 2022). With the progression of deep learning, these methods have been largely replaced in the following by Recurrent Neural Networks or Convolutional Neural Networks which automatically learn the relevant features for a correct classification (Do, 2018; Sadr et al., 2021). Pretrained Transformer-based LLMs have improved the performance of stance classification models even further (Gera & Neal, 2025; Mughal et al., 2024). Importantly, both encoder-only as well as generative models are able to be applied to the task.

The evaluation of stance classification models is typically performed using standard metrics such as precision, recall, and F_1 -score, similar to the evaluation of sequence labeling models described in Section 2.2.2. However, stance classification models do not operate on entity spans, but predict a single stance label for each target entity. A prediction is considered a TP if it matches the ground-truth label, and a FP if it does not. Conversely, a FN occurs when a ground-truth stance label is not correctly predicted by the model. These metrics are computed independently for each unique stance label present in the dataset. A standard approach to measuring overall model performance is to report macro-averaged scores across all stance classes (Gera & Neal, 2025). For a given entity, assuming a stance classification model predicts whether the speaker’s stance toward the entity is positive, neutral, or negative, the macro-averaged F_1 -score is computed as follows:

$$\text{Macro-Avg. } F_1 = \frac{F_1(\text{Positive}) + F_1(\text{Neutral}) + F_1(\text{Negative})}{3} \quad (2.8)$$

This measure gives equal weight to all stance classes, as the average is computed independently of the number of observations in each class. It is particularly suitable for imbalanced datasets, since the performance of the model on underrepresented classes is accounted for just as much as on overrepresented classes.

2.3 Contrastive Learning

Applying sequence labeling and stance classification to the task of detecting social group appeals enables the automatic identification of both the presence of a social group mention and the stance expressed toward that group. However, these approaches do not yet provide any information about the exact social group category that is referenced. To obtain such a categorization, one can leverage the distributed embedding representations of the group mentions. A way to enhance the semantic information of these vector representations is through the application of contrastive learning.

Contrastive learning is a subfield of representation learning whose goal is to learn a mapping from the input representation of a data point to an output feature vector. This output vector is intended to capture a meaningful representation of the data point. Deep learning models learn these representations automatically by optimizing an objective function. Contrastive learning, in particular, learns the mapping by comparing data points with respect to their similarity and

dissimilarity. It seeks a representation in which similar data points are positioned close together in the embedding space, while dissimilar data points are located far apart. The distance between data points is usually measured by computing the Euclidean distance. The learned mapping may function as a dimensionality reduction technique that generalizes well to unseen data points (Hadsell et al., 2006), as a preparatory step to provide input features for downstream applications such as supervised classification (Weinberger & Saul, 2009) or unsupervised clustering (Zhang et al., 2021). In the field of NLP, contrastive learning has been prominently applied in the development of SentenceBERT (Reimers & Gurevych, 2019) which is based on the classical BERT architecture and trained to produce an embedding vector that is a meaningful representation of the entire sentence.

The learning process can operate in either a self-supervised or supervised manner. In the self-supervised variant, no external information about the data points is required. Typically, data points are augmented to create similar but slightly modified instances of the same object. Every original data point is thus paired with a corresponding augmented instance, which inherently belongs to the same class. The model is then trained to map these paired instances to nearby locations in the output vector space, while simultaneously pushing the representations of unrelated data points farther apart. In supervised contrastive learning, the model leverages external ground-truth class information. In this setting, augmentations are not necessary, as the model is trained to bring the feature vectors of all data points from the same class closer together in the embedding space (Khosla et al., 2020).

Concerning the loss function used to learn the desired mapping, a distinction can be made between pair-based margin losses and probabilistic loss functions such as those based on Noise Contrastive Estimation (NCE) (Le-Khac et al., 2020). For pair-based margin losses, the model is provided with pairs of data points. It is trained to minimize the distance between positive pairs while ensuring that the distance between negative pairs exceeds a predefined margin. In contrast, NCE-based loss functions are designed to take a larger batch of negative pairs into account. Inspired by the probabilistic softmax classification objective, models trained with this approach aim to maximize the similarity between a positive pair, normalized by the sum of similarities to all negative examples in the batch.

2.4 Clustering in Embedding Space

The categorization of social group appeals into qualitative categories without prior knowledge of which categories generally exist can be achieved by applying a clustering algorithm. In contrast to supervised classification, clustering is an unsupervised learning technique whose objective is to group data points into previously unknown clusters without relying on labeled training data. Instead, cluster assignments are inferred directly from the data under the assumption that the underlying representations encode a latent structure that can be uncovered from their numerical properties (Jain et al., 1999). More specifically, clustering algorithms operate by computing the distances between data points. Based on these distances, the algorithms aim to form clusters in which observations are close to one another, while maximizing the separation between observations belonging to different clusters.

There exist a variety of clustering algorithms that differ primarily in their underlying assumptions and optimization strategies. A common distinction is made between hierarchical and partitional approaches. Hierarchical clustering algorithms initially assign each data point to its own cluster and then iteratively merge clusters based on pairwise distances and a chosen linkage criterion. That produces a hierarchy of nested cluster merges which are summarized in a dendrogram. In contrast, partitional clustering algorithms assume a predefined fixed number of clusters and aim to find an optimal partition of the data that adheres to this constraint (Jain et al., 1999). Due to its simplicity and efficiency, one of the most prominent partitional clustering techniques is the k-means algorithm (MacQueen, 1967). The algorithm begins by randomly

initializing cluster centroids and assigning each data point to the nearest centroid. The centroids are then recomputed as the mean of the assigned data points, after which the assignment step is repeated. This iterative process continues until a convergence criterion is satisfied. This may be that the cluster assignments do not change anymore after recomputing the cluster centroid.

Clustering algorithms are commonly applied to vector-based text representations in order to identify meaningful groups of semantically similar documents. Such vector representations may be derived from classical approaches, such as tf-idf, or from LLMs that produce contextualized embeddings. To make distance measures independent of vector magnitude, the embeddings are commonly normalized by their Euclidean norm. Documents with similar representations are consequently expected to be assigned to the same cluster. When used as input to clustering algorithms, embeddings generated by LLMs have been shown to produce more semantically coherent cluster assignments than classical text representations (Petukhova et al., 2025). Moreover, embeddings learned through contrastive learning are particularly well suited for clustering. Optimizing a contrastive loss encourages representations of similar data points to be positioned close together while dissimilar points are pushed apart, which preferably results in a uniform feature distribution on the hypersphere (T. Wang & Isola, 2020).

The effectiveness of a clustering algorithm in producing meaningful cluster assignments can be evaluated using internal or external clustering validation metrics. Internal metrics assess the clustering quality based solely on the data representation, whereas external ones rely on prior knowledge about the data to evaluate how well the resulting clusters reflect ground-truth groupings (Rendón et al., 2011). Since many partitional clustering algorithms require the number of clusters to be specified in advance, the validation metrics can be used as guidance for the selection of an appropriate number of clusters. An example of an internal validation metric is the silhouette score. For a given data point i , the silhouette score $s(i)$ is defined as follows:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \quad (2.9)$$

Here, $a(i)$ denotes the average distance between a specific data point i and all other data points assigned to the same cluster, while $b(i)$ represents the average distance between i and all data points in the nearest neighboring cluster. The silhouette coefficient ranges from -1 to 1 , with higher values indicating better clustering quality. By averaging $s(i)$ over all data points, an overall measure of clustering performance can be obtained.

An external evaluation of clustering quality can be performed using the Normalized Mutual Information (NMI) criterion. Mutual Information (MI), denoted as I , measures the agreement between two label assignments by quantifying how much information about one assignment is gained by knowing the other. Formally, the MI between two discrete random variables X and Y can be calculated by taking both the joint and marginal probabilities of all data points within X and Y into account.

$$I(X, Y) = \sum_{x \in X} \sum_{y \in Y} p(x, y) \log \frac{p(x, y)}{p(x)p(y)} \quad (2.10)$$

Since the raw MI-score is unbounded and depends on the entropy of the individual variables, it is commonly normalized to allow for meaningful comparison across different settings. A widely used normalization divides the MI by the geometric mean of the entropy H of both variables. The resulting NMI-score is then given by:

$$\text{NMI}(X, Y) = \frac{I(X, Y)}{\sqrt{H(X)H(Y)}} \quad (2.11)$$

The score is bounded between 0 and 1, where 0 indicates no statistical dependence between the two label assignments and 1 denotes perfect agreement. Consequently, NMI provides a measure of how well the cluster assignments Y reproduce prior knowledge about the data as

represented by the reference labels X .

Having laid out the foundations for classifying appeals to social groups, I now turn to review studies that are directly relevant to the methodological and substantive focus of this thesis.

Chapter 3

Related Work

The concept of social group appeals is an evolving research field and has been explored by scholars in a variety of ways. In this chapter, I review the literature on the operationalization of social group appeals as well as the methodological approaches that have been used to measure them. Furthermore, I outline the literature concerning the role of parliamentary speechmaking as a vote-seeking strategy for political parties at the collective and for legislators at the individual level. Finally, I illustrate the development of the relevance of class politics in the UK with the rise of voter-age as a reliable predictor for electoral outcomes.

3.1 Concept of Social Group Appeals

Social group membership has always been regarded as a decisive explanatory factor for shaping peoples' political opinion and electoral behaviour within the political science literature (Campbell et al., 1960; Nelson & Kinder, 1996). According to social identity theory, people tend to see themselves as members of particular social groups to which they have positive feelings, while separating themselves from perceived out-groups associated with negative emotions (Tajfel, 1974, 1981; Tajfel & Turner, 1979). When making political decisions, voters may inherently rely on this self-categorization as intuitive guidance (Kinder & Kam, 2010). Especially in complicated decision-making scenarios the social group membership may function as a cognitive shortcut by evaluating the political parties based on the social groups they represent most closely (Converse, 1964; Thau, 2019). Since talking about social groups allows parties to signal which groups are important to them (Conover, 1988), political actors can actively shape peoples' voting decision by targeting their social identity. Robison et al. (2021) experimentally show that symbolic appeals targeted at peoples' economic class directly influence their evaluation of political candidates. Rhetoric directed at the working-class leads to much better evaluations among working-class people whereas individuals who consider themselves as middle-class barely react to such statements.

Despite the widespread notion of social group membership's significance for peoples' political opinion, the study of how political parties strategically target social groups by referencing them is a relatively recent study field. It has been mainly established by Thau (2019) who posits that political parties use social group appeals deliberately to win votes and measures the use of these references in British party manifestos over a period of 50 years. He challenges the notion that social groups have stopped structuring electoral outcomes (Thomassen, 2005) and that parties have become dealigned from their core voters whose support they cannot take for granted anymore (Wattenberg, 2000). Instead, he demonstrates that both major British parties (Labour and Conservatives) have made steady use of group appeals while having become more diverse in the exact social groups they target. He defines a social group appeal as an "explicit statement that links a political party to some category of people" (Thau, 2019). While some scholars narrowly define a social group as being based on common socioeconomic or sociodemographic

traits (Huber, 2022), others are also considering shared perspectives or life experiences as uniting features (Howe et al., 2022; Licht & Szczepanski, 2025). Huber and Dolinsky (2023) outline the difficulty of unambiguously defining a social group and illustrate how the concept has been inconsistently applied in previous studies. According to them, most confusion arises around the question whether institutions or groupings based on political opinion should be regarded a social group. To provide a unified definition, the authors define a social group as a collective of people sharing sociodemographic traits. Further confusion arises between the conceptual difference of a social group mention and that of a social group appeal. When former studies only capture the mere reference to a social group and call it a mention (Thau, 2019, 2021), Huber and Dolinsky (2023) expand this concept by arguing that what is of interest are actually group appeals. These are intentional acts that associate a political actor with a social group. This association can be made in direct or indirect ways and in a broad variety of content domains. An example for an indirect appeal would be to appeal to a specific group by negatively mentioning a presumable out-group, in-group members have negative feelings against. To accurately capture the positional tone of the appeal, Horne et al. (2024) propose to additionally measure whether the author makes a positive, negative or rather neutral association with the group in question.

Another distinction needs to be drawn between policy-based and group-based appeals. So far, the literature on electoral competition has mainly focused on policy appeals (Dolezal et al., 2014). These are statements promoting specific policies which political actors hope to resonate well within their voter base and increases their support among these voters. Social group appeals on the other hand, function on a rather symbolic level aimed at the just described self-identification of people with a larger segment of society (Tajfel, 1974). However, as Thau (2019) states these concepts do not strictly need to stand in competition to one another. Instead, they might be used in combination to reinforce each other. To provide a more nuanced picture, many studies have started to measure explicitly if a reference to a social group also contains a mention of a specific policy (Horn et al., 2021; Horne et al., 2024; Riethmüller & Franzmann, 2025). Huber and Haselmayer (2025) provide a substantial explanation for how parties use policy- and group-appeals in combination. Inspired by issue ownership theory, they hypothesize that parties employ more appeals to groups when talking about policy issues they own and which are generally deemed important to their voters. Moreover, when a party’s core issue lacks salience, it might try to link it with substantively connected social groups with the hope of increasing its prominence. Riethmüller and Franzmann (2025) regard party status as a relevant factor in explaining to what degree a political actor talks about specific policies instead of social groups as such. By analyzing social media messages, they find that mainstream parties rely more on their policy record when appealing to groups. Challenger parties on the other hand often do not have such a distinct policy profile and are consequently forced to employ symbolic group appeals more often. Huber (2022) introduces the concept of a social group yield. She expects that politicians talk more about groups which are popular within the general electorate and perceived favorably among the party’s core voters. Moreover, it would be beneficial to attack those people who are rather unpopular across all voters and not liked by their own electorate.

Besides examining when it might be beneficial for a political actor, the literature has also shed light on the question if this rhetoric is actually effective in influencing the voters’ electoral decision. Survey experiments demonstrate that voters rate candidates and policy statements more positively when they are exposed to references to their social identity (Huber et al., 2024; Robison et al., 2021; Serra, 2025). Furthermore, Thau (2021) illustrates the positive relationship between the UK Labour Party’s vote share among working-class people and their previously employed level of group appeals to this voter group in their party manifesto.

3.2 Methods for Detecting Social Group Appeals

To analyze the content, motivations, or target groups of social group appeals, researchers need a robust method to measure them at first. The vast majority of authors rely here on a manual content analysis (Dolinsky, 2023; Horn et al., 2021; Howe et al., 2022; Huber, 2022; Thau, 2019, 2021). Focusing predominantly on manifesto data, they classify the texts by hand according to whether and where it contains an appeal to a collective which fits to their prior definition of a social group. Additionally, some authors also qualitatively assign each reference a group category to measure not only the appeal as such, but also the target group more directly (Huber, 2022; Thau, 2019). Thau (2019) does that not with a pre-defined list of group categories, but builds this categorization inductively based on a post-analysis of the already identified mention spans.

While manual content analysis has the advantage of high validity since a human oversees all texts, it has the major drawback of being very resource-intensive which naturally limits the researcher to a smaller corpus. The natural advancement to this are dictionary-based approaches in which the researcher defines a list of keywords that are then applied to all texts to treat all matches as predictions for a group mention. For the detection of social group appeals, this process is only applied by Riethmüller and Franzmann (2025) who create a dictionary to automatically detect group references in tweets. They divide their keywords into 24 group categories to also measure the respective target group. Moreover, they also classify the sentiment toward the targeted group, but rely on a supervised machine learning classification model (Hamborg & Donnay, 2021) to do so.

One reason for the relatively small number of published studies using dictionary-based methods to detect group appeals may be the growing success of LLMs for classification tasks, which have also gained prominence in the social sciences. Social groups can be referenced in a wide variety of forms that also vary considerably in length, which makes it challenging for a dictionary-based approach to capture them exhaustively. (Licht & Szczepanski, 2025). LLMs however are, due to their rich semantic understanding of text, still able to accurately detect these mentions without requiring the researcher to manually define a long list of keywords. These models have already been successfully applied to the detection of group appeals by fine-tuning encoder-based DeBERTa models on a manually annotated training set (Horne et al., 2024; Licht & Szczepanski, 2025). Horne et al. (2024) do not only detect the mention span itself, but also the position taken towards the group, whether the sentence contains a policy reference and the qualitative group category. They achieve this by fine-tuning several distinct models on each specific task to then use them in a pipeline.

To this point, the majority of scholars applying LLMs have relied on model variants of the BERT family. Nonetheless, Jalali Farahani et al. (2024) show that it is possible to apply generative LLMs to the task. They claim that no supervised fine-tuning is needed for these models to detect group appeals, but that they can be prompted with some few-shot examples and achieve similarly good results as a BERT model. They force the LLM to predict the social group mentions surrounded by special tokens (@ for the start and # for the end of a span). This procedure has been first suggested by S. Wang et al. (2025) and has the advantage of framing the classification task as a text generation exercise while still allowing to construct mention spans in a similar fashion as it is done with BIO tags. So far, there does not exist any study that uses a generative LLM to predict the author’s stance towards the mentioned group. However, existing studies highlight the possibilities that generative LLMs provide for automated stance classification via in-context learning, CoT prompting or parameter-efficient finetuning (Gera & Neal, 2025; Mughal et al., 2024).

Except for Thau (2019) there are no studies yet which apply an inductive approach to the categorization of group references into qualitative group categories. Either, this categorization process is not done at all (Dolinsky & Huber, 2023; Licht & Szczepanski, 2025) or it is done based on supervised classification into pre-defined group categories (Horne et al., 2024; Riethmüller

& Franzmann, 2025). Recent work in the field of open-named entity recognition points to the possibilities of applying unsupervised clustering methods to the embedding representation of the entity span in order to derive a meaningful group categorization the researcher can then manually label or prompt a generative LLM to provide substantive category names (Genest et al., 2025). As Licht and Szczepanski (2025) argue, it is essential to further develop inductive classification approaches. The main reason is that it is difficult to anticipate all possible group categories in advance.

3.3 Vote-Seeking Through Parliamentary Speechmaking

According to Strøm (1997), legislators’ behavior in parliament is shaped by several overarching goals. The most important one of them is securing the personal reelection since it serves as a prerequisite for achieving further ambitions such as obtaining certain executive or leadership positions. Parliamentary speeches constitute an important instrument for MPs’ attempts to maximize reelection prospects, as they allow them to communicate their policy positions to their own party, to competing parties, and most importantly directly to their voters (Proksch & Slapin, 2012). When delivering a speech, legislators may either adhere closely to the party line or seek to cultivate a distinct personal profile. Carey (2007) speaks of competing principals that may push the respective legislator in different directions. Speechmaking can therefore function as a vehicle for signaling varying degrees of dissent from the party leadership (Herzog & Benoit, 2015). This behavior proves to represent a promising vote-winning strategy for an individual MP, but most often turns out to be costly for the party as a whole (Kam, 2009).

There is broad agreement that the extent to which legislators prioritize party conformity over personal reputation depends primarily on the country’s institutional setting (Carey, 2007; Carey & Shugart, 1995; Crisp et al., 2004; Proksch & Slapin, 2012). With regard to parliamentary speechmaking, Proksch and Slapin (2012, 2015) argue that legislative speeches are shaped by strategic considerations within parties. They conceptualize speechmaking as a delegation game in which the party leader decides whether to give a speech herself or to delegate it to a backbench MP. This decision depends on how important a unified party message is under the given electoral system. Consistent with their theoretical expectations, they show that in Germany, where institutional arrangements promote strong party unity, party leaders take the floor significantly more often compared to their colleagues. In contrast, in the UK, whose single-member district system creates stronger incentives for individual differentiation, backbenchers are more frequently permitted to speak. Other scholars contend that the effects of electoral institutions are mediated by legislators’ individual electoral vulnerability (André et al., 2015; Kellermann, 2016). MPs who secured their seat by a narrow margin face stronger incentives to cultivate a personal profile and signal responsiveness to their constituency. Supporting this argument, Kellermann (2016) finds that electorally vulnerable British MPs systematically ask more parliamentary questions than colleagues who hold relatively safe seats.

An important component of cultivating a personal vote is establishing a visible connection to the local constituency and demonstrating responsiveness to constituents’ concerns. Mayhew (2004) analyzes this vote-seeking behavior in the United States and documents substantial levels of constituency service aimed at building what he terms an “electoral connection”. Constituency-oriented behavior may take the form of parliamentary speeches in which MPs articulate their ideological positions or highlight personal achievements that benefit their local district. Although the UK is generally characterized as more party-centered, similar patterns of constituency work can be observed there (Cain et al., 1984, 1987).

3.4 Class Politics in the UK

For much of the post-war period, British politics was structured around economic class as the central organizing principle of both party competition and voter behavior. The working class constituted the electoral core of the Labour Party, which mobilized support through programmatic appeals rooted in labor representation and redistribution. The Conservative Party, on the other hand, relied more heavily on cross-class coalitions, while maintaining particularly strong ties to business interests (Ford & Goodwin, 2017). In this respect, the British party system closely reflected the expectations of classical cleavage theory as formulated by Lipset and Rokkan (1967). That says that party systems become structured around long-lasting social division lines such as the conflict between workers and employers. These cleavages generate stable patterns of political identification and produce alignments between social groups and parties. Over time, however, the structuring force of class has weakened. Processes of social and economic transformation have contributed to a gradual erosion of traditional partisan loyalties, as voters have become increasingly dealigned from the parties they once consistently supported (Dalton, 2018; Mair, 2023). As a result, social position no longer determines vote choice to the same extent as in earlier decades, and the link between occupational class and party preference has weakened considerably. This dealignment has also contributed to a more fragmented and competitive party system, in which the electoral dominance of Labour and the Conservatives has become less secure (Dunleavy, 2005). In the UK, the loosening of ties to the working-class electorate became particularly visible for the Labour Party during the 1990s, when the party substantially reoriented its electoral strategy. Under the leadership of Tony Blair and the banner of “New Labour”, the party deliberately repositioned itself toward the political center, seeking to broaden its appeal beyond its traditional working-class base and to attract middle-class voters, with a particular focus on university graduates (Ford & Goodwin, 2017). This strategic shift was not merely rhetorical but reflected a broader transformation in how Labour conceptualized its electoral coalition. Empirical evidence supports this interpretation. Thau (2019) documents a marked decline in explicit appeals to the working class in Labour’s party manifestos during this period. Strikingly, he finds that by the late 1990s, the two major parties had become largely indistinguishable in their references to working-class groups. Instead of emphasizing economic constituencies, both Labour and the Conservative Party increasingly target socially neutral categories, such as families and parents. These findings align closely with Evans and Tilley (2017), who show that both major parties shifted their communication toward broad, non-conflictual social groups in order to avoid signaling overt alignment with, or opposition to, specific socio-economic constituencies. This communicative convergence was accompanied by a substantive convergence in left–right policy positioning of the two parties. Sobolewska and Ford (2020) and Ford and Goodwin (2017) argue that the convergence of Labour and the Conservative Party around a liberal consensus left parts of the predominantly white and lower-educated electorate politically unrepresented. This created opportunities for challenger parties such as the UK Independence Party, which combined Euroscepticism with a strong anti-immigration platform and increased pressure on the party system (Evans & Tilley, 2017). These developments reflect the emergence of a transnational cleavage structured around European integration and immigration (Hooghe & Marks, 2018), ultimately culminating in the Conservative Party’s pledge to hold a referendum on UK’s membership in the European Union.

While economic class has lost much of its explanatory power in accounting for electoral behavior in the UK, other demographic variables have gained prominence. Most notably, age has emerged as one of the strongest predictors of vote choice. As Serra (2024) demonstrates, although age has long been associated with voting behavior in Britain, its predictive strength has increased markedly since 2010. Younger voters are now substantially more likely to support the Labour Party, whereas older voters increasingly align with the Conservatives. One common explanation attributes this divergence to a growing ideological polarization between younger and older cohorts. Yet more recent research challenges this interpretation. O’grady (2023),

shows that ideological differences between age groups have long been present and may even be narrowing rather than widening. Instead, Serra (2024) argues that the expanding age gap rather reflects a rise in the salience of progressive and conservative ideologies which consequently increasingly shape peoples' vote decisions. This interpretation resonates with broader accounts of the rise of identity politics, in which cultural and value-based considerations increasingly structure electoral decision-making and replace the class-based alignments that once dominated British politics (Sobolewska & Ford, 2020). Beyond value salience, Serra (2024) states that key life events traditionally associated with more conservative attitudes, such as marriage and homeownership, occur later in life for younger generations than for previous cohorts. As a result, the attitudinal shifts that once accompanied early adulthood now take place at later stages, prolonging the period during which young voters lean toward Labour. She also emphasizes the strategic role of parties themselves. In a survey experiment, Serra (2025) examines how different age groups respond to substantive and symbolic appeals targeted specifically at their age cohort. The findings indicate that younger voters exhibit significantly higher levels of support for parties that explicitly advocate their generational interests, both in substantive and symbolic terms. Crucially, appeals directed at young voters do not necessarily reduce support among old voters. This suggests that there is strategic room for the Conservative Party to narrow the existing age gap by engaging young voters without alienating its established electoral base.

In this chapter, I reviewed the most relevant studies on the concept and quantification of social group appeals. I also discussed the role of parliamentary speechmaking as a vote-seeking strategy and examined the declining influence of economic class in British politics alongside the increasing structural significance of age cohorts. I continue by outlining the methodology used to develop and evaluate the classification and clustering models for capturing social group appeals.

Chapter 4

Methodology of Model Development

I address the detection of social group appeals in political text by developing a pipeline of classification and clustering models. In the following chapter, I provide a conceptual definition of social group appeals and outline how I measure this concept by training and evaluating different LLM models alongside a dictionary-based baseline.

4.1 Definitions and Data Sources

Before training any classification model to detect social group appeals, it is essential to define the general concept precisely. This is done in the following section, in which I also describe the data sources used to train the classification and clustering models.

4.1.1 Definition of a Social Group Appeal

The literature offers several competing definitions of what constitutes a social group and how social group appeals should be conceptualized. While Huber and Dolinsky (2023) define social groups on the basis of shared or acquired sociodemographic traits, other scholars adopt broader understandings that also encompass shared life experiences or common values (Howe et al., 2022; Licht & Sczepanski, 2025). I adopt the narrower definition proposed by Huber and Dolinsky (2023) and conceptualize a social group as a segment of society characterized by shared or acquired sociodemographic attributes. Such attributes include, for example, gender, age, ethnicity, language, income, or place of residence. Following this approach, I do not regard institutionalized collective actors such as trade unions or the police as social groups. However, I include references to specific people within these institutions, such as mentions of union members or police officers. This corresponds to the idea that social group appeals are directed at people rather than organizations (Huber & Dolinsky, 2023). I further exclude groupings defined solely by shared political opinions. This comparatively restrictive conceptualization is chosen to enhance conceptual clarity. Given the already blurred boundaries of the concept, a narrower definition should help to avoid ambiguity and to ensure a more consistent detection of social group appeals.

I treat any general reference to a social group as a group mention. By additionally measuring the positional tone a speaker adopts toward the referenced group, I capture the concept of a group appeal. Following Huber and Dolinsky (2023), I define a social group appeal as an intentional act that associates a political actor with a social group in either a supportive, critical, or neutral manner. In line with this definition, I focus exclusively on explicit statements made by British legislators in parliamentary speeches. A positive stance toward a social group is given when the speaker expresses support for that group. Such support may be articulated explicitly, but it may also appear in more subtle forms. For instance, beyond direct statements of endorsement, raising the group’s interests is also treated as a supportive act. This may occur if a political actor poses

a question that highlights the group’s rights or concerns without overtly stating that he or she wants to support the group. Positive appeals may further include specific policy commitments or symbolic expressions of solidarity with the group. A negative stance is present when a political actor criticizes a social group directly or indirectly, or otherwise conveys negative sentiments toward it. Importantly, in such cases the negatively referenced group is not necessarily the group being appealed to. Rather, the appeal may target another group that shares opposition to the criticized group. This corresponds to what Huber and Dolinsky (2023) describe as an indirect group appeal. A neutral stance is recorded when the speaker neither expresses support nor criticism. This may occur, for example, when the group is mentioned in a purely factual or descriptive manner. More detailed definitions of social groups and social group appeals are provided in Appendix A. There, I also discuss ambiguous edge cases that are not directly covered by the definition outlined here and explain how they are handled in the classification process.

4.1.2 Expert Dictionaries

In this work, I develop LLM-based classification models for the detection of social group appeals. However, it is essential to compare the performance of the LLMs with a method that requires less computational resources. Using a dictionary for the classification of group appeals provides such a baseline as it only requires setting up a list of keywords that best capture the concept one wants to measure. For this purpose, I am using the dictionary created by Horne et al. (2024) who are experts in this field and granted me permission to use their work as part of this thesis. Hence, using their dictionary as a baseline method represents a best-effort to automatically detect social group references without the need to train a more complex model. It is particularly well suited as a comparative approach since my definition of a social group appeal is largely inspired by Huber and Dolinsky (2023) who are also the authors of the dictionary.

Applying this dictionary for keyword-matching provides a means to detect spans of social group mentions. Moreover, as it groups all keywords into 45 social group categories, it allows to directly assign each mention a qualitative category. However, this does not make it possible to classify the speaker’s stance toward the referenced group. For this reason, I am using the 2015 Lexicoder Sentiment Dictionary (LSD) developed in its original version by Young and Soroka (2012) and freely available as part of the R-package *quanteda* (Benoit et al., 2018). A detailed description about the exact application of both these dictionaries can be found in the following sections.

4.1.3 Annotation Dataset and Data Augmentations

To obtain the full record of parliamentary speeches given in the UK House of Commons, I rely on the second version of the ParlSpeech dataset (Rauh & Schwalbach, 2020), which contains all speeches given in the British parliament between 1988 and 2019. To train and evaluate the classification models, I rely on a manually compiled dataset of 5,000 sentences drawn from parliamentary speeches delivered during question debates in the British Parliament between May 6, 2010, and December 12, 2019. The dataset consists of two equally sized subsets. The first comprises sentences containing at least one phrase that matches a social group expression listed in the expert dictionary developed by Horne et al. (2024). The second consists of randomly sampled sentences from the same type of debates and time period that do not match any dictionary entry. Many dictionary entries contain asterisks to allow for flexible matching, including additional prefixes, suffixes, or entire words at specific positions within a phrase. To implement this flexibility, I construct a regular expression pattern that combines all unique phrases using logical OR operators and translates the asterisks into wildcard expressions that permit additional characters or words before or after specified terms. Applying this regular expression to the full set of speeches yields two pools of sentences. The first includes all sentences that match at least one dictionary entry, while the second contains those sentences that do not match. From

each pool, I draw 2,500 sentences using stratified random sampling based on the speaker’s political party and the year of the speech to ensure a balanced representation across parties and time. After compiling the dataset, I manually annotate each sentence for the presence of social group appeals, closely following the definition of a social group appeal and several annotation guidelines I outline in the codebook which is included in Appendix A. The annotation process is conducted using a locally hosted version of Label Studio (Tkachenko et al., 2020). For each sentence, I identify and mark the textual span corresponding to a social group reference and classify the speaker’s stance toward the group as either positive, neutral, or negative. A sentence may contain no appeal, a single appeal, or multiple appeals. Each appeal may consist of one or several words. Label Studio then allows all annotations to be exported as a JSON file, in which the annotated spans are listed according to their start and end character indices within the sentence.

To determine an appropriate dataset size, I monitor the performance of a BERT model while incrementally increasing the number of training observations. This allows me to assess the point at which performance gains begin to plateau. In addition, I generate two augmented versions of each sentence to synthetically expand the dataset without requiring further manual annotation. For the first augmentation strategy, I randomly replace 30% of all tokens that are not part of a labeled social group appeal span with semantically similar alternatives. To identify suitable replacements, I use a pre-trained GloVe embedding model and compute cosine similarity scores to retrieve the five most similar words for each candidate word. The original word is explicitly excluded from this set. From the remaining five candidates, one replacement term is randomly sampled. I update the index positions of each group mention after augmentation to ensure accurate span alignment. For the second augmentation strategy, I use OpenAI’s generative GPT-5-nano model via API requests to produce a complete paraphrase of the original sentence while keeping all social group references unchanged. If a sentence does not contain any group references, the model is prompted to generate a full paraphrase without further constraints. Subsequently, I identify the exact position of each reference in the paraphrased text in order to compute the updated start and end indices. Additionally, I verify that all annotated group mentions from the original sentence are correctly preserved and detectable in the paraphrased version. To evaluate the effectiveness of the augmentation strategies, I empirically assess whether the inclusion of augmentations in the training data improves the classification performance of an encoder model.

The final annotated dataset comprises 5,000 original sentences and 10,000 augmented sentences, resulting in a total of 15,000 observations. However, not all sentences contain a social group appeal and are therefore suitable for training the stance classification model. To address this, I construct a subset that includes only instances in which a stance toward a social group is expressed. Because a single sentence may contain multiple social group appeals, I treat each appeal as a separate training instance for the stance classification models. For both classification tasks I create a training set containing 80% of the observations and a validation set that contains the remaining 20%. The splits are created by random sampling. For the stance classification task I apply stratified random sampling to preserve the overall proportion of stance labels in both splits. Crucially, for all data subsets I ensure that no augmented sentence is part of the training data, when its original counterpart is included in the validation set. Otherwise, the information from the augmented version would leak into the validation set and inflate model performance artificially.

4.2 Sequence Labeling Model

In accordance with the principles of sequence labeling, I create classification models that predict a sequence of output labels whose length is identical to the length of the input sequence. Thus, for each word in a sentence, the models predict a label that indicates whether or not it is part

of the mention of a social group. To do this, I rely on the BIO tagging scheme (Ramshaw & Marcus, 1995; Tjong Kim Sang & De Meulder, 2003). Here, the model should predict the “B-tag” if a word represents the beginning of a mention of a social group, the “I-tag” if it is the continuation of an ongoing mention, and the “O-tag” if the word is outside of any mention. This tagging scheme enables to cleanly separate the boundaries between directly adjacent group mentions. In principle, the tagging scheme allows for the annotation of multiple entity types, each indicated by a distinct label following the BIO convention. For my application however only the entity type of social groups is relevant. Accordingly, the model is trained to predict three tags: “B-sg”, “I-sg” and “O”.

I implement several BERT-based models for the detection of social group mentions in order to assess which encoder architecture is best suited for the task. Specifically, I evaluate the original BERT model (Devlin et al., 2019), its optimized variant RoBERTa (Liu et al., 2019), the computationally lighter DistilBERT (Sanh et al., 2020), and DeBERTa, which works with a disentangled attention mechanism (He et al., 2021). I use the base versions of all these models. I fine-tune them with the annotated sentences contained in the training set. All models are fine-tuned by adding a task-specific token classification layer that predicts one label per token according to the BIO tagging scheme. The cross-entropy loss is applied to measure the models’ performance and update their parameters. Because the task requires token-level predictions, I assign each token a ground-truth label derived from the manual span annotations. All BERT-based models rely on subword tokenization (Kudo & Richardson, 2018; Sennrich et al., 2016; Wu et al., 2016), meaning that a token does not necessarily correspond to a complete word. Consequently, a single word-level BIO tag may be represented by several token-level tags. Because the objective is to obtain predictions at the word level, the token-level outputs must be mapped back to words after model inference. The mapping between subtokens and their corresponding words can be recovered using the word IDs provided by the tokenizer associated with each BERT model. To perform this reversion, I define a set of deterministic aggregation rules that collapse subtoken predictions into a single word-level label. If a word contains both “B-sg” and “I-sg” predictions across its subtokens, the word is assigned the label “B-sg”. If all subtokens of a word are labeled “I-sg”, the word receives the label “I-sg”. If any subtoken of a word is assigned the label “O”, the word is classified as “O”. This aggregation strategy is intentionally conservative. An alternative approach would be to apply majority voting across subtokens.

For each BERT model, I tune a set of hyperparameters using the held-out validation set. Specifically, I optimize the learning rate, weight decay, and batch size. For each hyperparameter, I define three candidate values that are shown in Table 4.1, resulting in 27 total possible combinations. The selection of hyperparameter ranges is guided by Licht and Sczepanski (2025), who likewise conduct hyperparameter tuning on BERT models for social group mention detection. To efficiently search this space, I employ the Tree-Structured Parzen Estimator (TPE) algorithm as implemented in the Python package Optuna (Akiba et al., 2019). I conduct 20 optimization trials per model. In each trial, the model is trained for up to ten epochs, and the F_1 -score on the validation set is computed after every epoch. If the validation performance does not improve for three consecutive epochs, training is terminated early. The epoch corresponding to the highest validation F_1 -score is retained as the optimal stopping point. This procedure not only identifies the best-performing hyperparameter configuration but also implicitly tunes the number of training epochs through early stopping.

As an alternative to the BERT models, I evaluate generative LLMs for their suitability in detecting social group mentions. Specifically, I assess the performance of OpenAI’s GPT-4o-mini and GPT-5-nano models. Both are applied using a few-shot prompting approach. Rather than fine-tuning model parameters on the task, I provide a task explanation as well as a set of few-shot examples written in Markdown syntax within the input prompt to guide the model’s behavior. To evaluate the sensitivity of model performance to prompt design, I construct prompts of

Hyperparameter	Value 1	Value 2	Value 3
Learning rate	$9e^{-06}$	$2e^{-05}$	$4e^{-05}$
Weight decay	0.01	0.1	0.3
Batch size	8	16	32

Table 4.1: Hyperparameter search space for all BERT models in both classification tasks. Optimal hyperparameters are searched for 20 trials with the TPE-algorithm implemented within the Optuna package (Akiba et al., 2019) via application to the validation set.

varying levels of detail and with different numbers of few-shot examples. An example of the least detailed prompt used in this study, with a set of few-shot examples and a test sentence, is shown in Figure 4.1. All designed model prompts are included in Appendix B. The model is instructed to return the detected mentions in JSON format. This structured output facilitates the systematic extraction of predicted spans. Based on these JSON files I convert each sentence to a sequence of word-level BIO tags. To achieve this, I identify all occurrences of each predicted mention span within the original sentence and map them onto the corresponding word positions.

```

System Message:
## Task Objective
Extract all qualifying social groups from the sentence as a JSON object. Do not
paraphrase the mentions at all. If no social groups are present, return an empty JSON
array.

## Definition of a Social Group
A social group is a collective of people sharing socio-demographic attributes
(age, ethnicity, income, occupation etc.).

Do not extract institutional groups, firms, state authorities, groups based on
political opinion or very specific small collectives such as one single family.
Do extract groupings of people within an institution and singular forms if it is
generalizing to a broader collective.
Extract only the social group component unless there is an
additional description to it.

User Message:
Sentence: We need to do more for young people, especially those with physical
disabilities.

Assistant Message:
{"social_groups": ["young people", "those with physical disabilities"]}

User Message:
Sentence: The Government is working in close cooperation with the Labour Party on this
matter.

Assistant Message:
{"social_groups": []}

User Message:
Sentence: Our party supports the rights of all young people living in this country.

```

Figure 4.1: Least detailed prompt for social group mention detection with two few-shot examples. Used as input for GPT-4o-mini and GPT-5-nano via OpenAI’s “Responses” API.

I access the closed-source generative LLMs via the OpenAI API. To reduce latency and improve throughput, I send multiple asynchronous requests while ensuring that the total request volume does not exceed the rate limits imposed per minute. In addition, I tune selected model-specific hyperparameters. For GPT-4o-mini, I vary the temperature parameter $\{0, 0.25, 0.5\}$, which controls the degree of randomness in the output distribution. For GPT-5-nano, I tune the reasoning effort parameter $\{\text{“None”}, \text{“low”}, \text{“medium”}, \text{“high”}\}$. This parameter is available only for GPT-5-nano, as GPT-4o-mini does not belong to the reasoning model family. Reasoning models are designed to internally generate intermediate reasoning steps before producing a final answer. That behavior is conceptually equivalent to the CoT approach, which encourages models to articulate intermediate reasoning to improve performance on complex tasks.

As a baseline to the two LLM-based approaches, I additionally implement a dictionary method for detecting social group mention spans. For this purpose, I rely on the expert dictionary compiled by Horne et al. (2024). Consistent with the procedure used to construct the training data, I perform keyword matching using a comprehensive regular expression that combines all social group keywords via logical OR operators. The detected spans are subsequently post-processed to assign BIO tags at the word level, ensuring comparability with the evaluation procedure applied to the other approaches. Since this method does not involve model training or probabilistic inference, no hyperparameter tuning is required.

I evaluate all approaches using the sequeval metric, which is outlined in the following. The evaluation is conducted by applying 5-fold cross-validation. I partition the training dataset into five equally sized folds. Across five iterations, I fine-tune each model using four folds for training and evaluate performance on the remaining held-out fold. All models get fine-tuned with the hyperparameters that got selected on the validation set before. For methods that do not require fine-tuning, evaluation is performed directly on the respective held-out fold without additional training. The final evaluation estimates are then computed by taking the average of the five individual performance estimates.

To evaluate the performance of the sequence labeling models, I apply three metrics of increasing strictness. First, I use the sequeval metric introduced by Ramshaw and Marcus (1995) and implemented as a Python package by Nakayama (2018). This metric adopts a strict span-level evaluation criterion. A predicted entity is counted as a TP only if its boundaries exactly match the corresponding ground-truth annotation. If the predicted span deviates in any way, it is counted as a FP, while the unmatched annotated span simultaneously contributes a FN. Based on the resulting counts of TPs, FPs, and FNs, the standard classification metrics precision, recall, and the F_1 -score are computed as defined in Equations 2.5, 2.6 and 2.7. Because the task involves only a single entity class, no macro-averaging across multiple classes is required.

Following Licht and Sczepanski (2025), I argue that sequeval may impose an overly strict requirement in my application context. Compared to outlining the boundaries of named entities, determining the precise boundaries of a social group mention is not always unambiguous. Models may omit preceding articles or descriptive modifiers that extend beyond the core group noun. From a substantive perspective, it is often more important that a model correctly identifies the general presence of a social group mention, even if minor boundary discrepancies occur. Therefore, in addition to strict span-level evaluation, I adopt the cross-span average metric proposed by Licht and Sczepanski (2025). Under the cross-span average metric precision, recall, and F_1 -scores are computed separately for each individual span in the combined set of ground-truth annotations and model predictions. For each span, the frequency of correctly predicted words (TPs), incorrectly predicted words (FPs), and missed words (FNs) is used to calculate the respective metric in accordance with the standard definitions of precision, recall and F_1 -score. The resulting span-level scores are then averaged across all spans to obtain cross-span scores for all metrics. This procedure leads to divisions by zero in two cases. If a model predicts a span for which none of the words overlap with a ground-truth span, the recall denominator equals zero. In such cases, recall is undefined and is therefore set to zero. An analogous situation arises

when a ground-truth span contains no correctly predicted words. Here, the denominator of the precision score is undefined. Consistent with the treatment of recall, precision is set to zero in these cases. This convention ensures that non-overlapping spans are penalized appropriately while maintaining numerical stability of the evaluation metrics. The functioning of the metric is demonstrated in Figure 4.2. As the metric itself, the illustration is inspired by Licht and Sczepanski (2025).

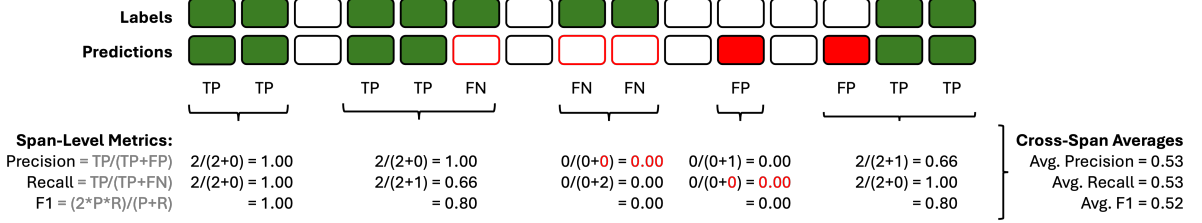


Figure 4.2: Illustration of mention-level evaluation metric adapted by Licht and Sczepanski (2025)

Finally, I introduce an additional evaluation metric, which I call “mention-detection”. While the cross-span average metric reduces the penalty for partial overlaps compared to sequeval, it still penalizes boundary mismatches. In some applications, however, the primary interest lies in detecting the mere presence of a social group mention rather than identifying its exact span boundaries. Under the mention-detection metric, a predicted span is counted as a TP if it overlaps with the corresponding ground-truth span by at least one word. If a predicted span does not share any word with a ground-truth annotation, it is treated as a FP. Conversely, if no word of an annotated span is recovered by any prediction, this instance is counted as a FN. Based on these counts, precision, recall, and the F_1 -score are computed.

4.3 Stance Classification Model

In accordance with the definition of a social group appeal, I develop classification models that predict, for each individual occurrence of a social group mention, whether the speaker expresses a negative, neutral, or positive stance toward that group. In analogy to the sequence labeling task, I implement stance classification using encoder-only LLMs, generative LLMs, and a computationally lightweight dictionary approach. For the encoder architecture, I develop two types of BERT-based stance classification models. The first follows a sequence classification setup. The input consists of the tokenized sentence and the respective social group reference, separated by the special separator token ([SEP]). If a sentence contains multiple social group mentions, I generate separate input instances for each mention by appending the respective group after the separator token. A linear classification layer is added on top of the contextualized [CLS] vector representation to predict the speaker’s stance toward the referenced group. Model parameters are optimized using cross-entropy loss. I implement this architecture using the same pretrained BERT models as for sequence labeling (BERT (Devlin et al., 2019), RoBERTa (Liu et al., 2019), DistilBERT (Sanh et al., 2020), and DeBERTa (He et al., 2021)). Hyperparameters are tuned on the held-out validation set using the TPE algorithm with 20 search trials, implemented via the Optuna framework. The search space corresponds to the hyperparameter configurations reported in Table 4.1. During training, early stopping is applied if validation performance does not improve for three consecutive epochs and the epoch delivering the best validation performance is stored.

The second BERT-based approach relies on NLI models. NLI models can determine whether a hypothesis logically follows from a given premise and can therefore be used for zero-shot classification tasks. Similar to standard sequence classification architectures, the premise and hypothesis are tokenized jointly and separated by the [SEP] token. I employ the NLI model introduced

by Laurer et al. (2024), which is based on a pretrained DeBERTa architecture fine-tuned on multiple NLI benchmark datasets as well as synthetically augmented training data. For each social group mention within a training sentence, I construct three premise–hypothesis pairs. The premise corresponds to the original sentence, while the hypothesis states that the sentence expresses either a positive, negative, or neutral stance toward the respective social group. The predicted stance is determined by selecting the hypothesis with the highest entailment probability. Because each manually annotated group mention has exactly one ground-truth stance label, two of the constructed premise–hypothesis pairs correspond to contradiction, while one represents entailment. Model training proceeds by minimizing the cross-entropy loss over the predicted entailment versus contradiction labels. I apply the same hyperparameter-tuning as for the sequence classification models.

The training dataset exhibits a substantial class imbalance, with positive stance instances occurring considerably more frequently than neutral or negative ones. To address this imbalance, I incorporate augmented versions of sentences with neutral and negative stance as additional training instances in order to oversample the minority classes during fine-tuning. I examine different oversampling strategies, including oversampling only the negative class, oversampling both the negative and neutral classes, and proportionally oversampling all classes. These strategies are evaluated on the performance of an NLI-BERT model.

For the generative LLMs, I follow the same procedure as in the social group detection task by applying a few-shot prompting approach to the GPT-4o-mini and GPT-5-nano models. In the system prompt, I provide detailed instructions specifying how to classify the stance toward a referenced social group, which stance categories are possible, and how these categories are defined. I then include few-shot examples covering all possible stance labels. To assess the effect of different prompt designs, I construct prompts with varying degrees of detail and experiment with different numbers of few-shot examples. I vary solely the number of sentences containing a positive stance, because I want to ensure that the model always receives at least one few-shot example from each stance category. As in the social group detection task, I tune model-specific hyperparameters on the validation set. For GPT-4o-mini, I vary the temperature parameter $\{0, 0.25, 0.5\}$, and for GPT-5-nano, I adjust the reasoning effort parameter $\{\text{“None”}, \text{“low”}, \text{“medium”}, \text{“high”}\}$. The full set of prompts used in the experiments is provided in Appendix B.

Furthermore, I assess the effectiveness of a dictionary baseline that relies on the bag-of-words approach. A fundamental limitation of bag-of-words models is that they disregard syntactic structure and treat all words independently of their relative positions. This makes the direct implementation of a dictionary-based stance classifier difficult since the task inherently requires to take the sentence structure into account. Consequently, I employ a sentiment analysis approach that assigns words to positive and negative lexical categories. For this purpose, I use the LSD (Young & Soroka, 2012), which has demonstrated strong performance in the analysis of political texts and has been shown to yield more robust results in this domain than alternatives such as the Linguistic Inquiry and Word Count dictionaries. Following Lowe et al. (2011), I compute a smoothed log ratio of positive to negative word counts.

$$S = \log \left(\frac{\text{pos}_i + 0.5}{\text{neg}_i + 0.5} \right) \quad (4.1)$$

Here, pos_i and neg_i denote the number of positive and negative words in sentence i , respectively. The constant 0.5 serves as a smoothing term to avoid division by zero and to shrink extreme values toward zero. The logarithmic transformation ensures symmetry of the score around zero. The resulting value S represents the overall sentiment of the sentence. To derive a stance classification for a specific social group mention, I extract words from a context window surrounding the mention span. The optimal window size is determined on the held-out validation set. In the same fashion, I tune a positive and a negative threshold that map the continuous sentiment score to discrete stance categories.

As in the sequence labeling task, all stance classification models are evaluated using 5-fold cross-validation on the training dataset. The folds are created via stratified random sampling to preserve the overall distribution of stance labels across splits. For the evaluation of all stance classification models, I compute precision, recall, and F_1 -score separately for each stance label and then aggregate these scores using the macro-average, as defined in Equation 2.8. Macro-averaging is particularly appropriate in this context due to the pronounced class imbalance, especially between the negative and positive labels. By assigning equal weight to each class, the macro-average ensures that model performance across all stance categories is considered, rather than being dominated by the majority class.

4.4 Categorization via Clustering and Contrastive Learning

The described classification models predict the presence and boundaries of social group mention spans, as well as the speaker’s stance toward the referenced group. However, they do not indicate which specific social group is being referenced. A possibility to achieve this would be to develop a separate classification model with as many class labels as the social group categories of interest. My objective, however, is to develop an inductive method that can identify qualitative social group categories in an unsupervised manner, without predefining the categories. To achieve this, I propose an unsupervised clustering approach that operates on semantically informative embedding vectors generated by a BERT model. To further refine the structure of the embedding space and enhance separability between categories, I incorporate contrastive learning techniques. A similar procedure has been suggested by Genest et al. (2025) for the task of open-world NER.

The social group mentions are encoded by prompting a BERT model. As an encoder architecture, BERT does not have the ability to generate text autoregressively. Instead, it is trained to predict the most likely token at a masked position. I leverage this MLM objective by inserting each social group mention sg into a fixed prompt template P containing the [MASK] token.

$$P(sg) = \text{“Social group of } sg \text{ is [MASK].”} \quad (4.2)$$

The vector representation of the [MASK] token contains semantic information about the social group category, as BERT is trained to predict the most likely term at this position. To obtain the final embedding used for clustering, this representation is passed through a linear projection layer that reduces its dimensionality to 128. This reduction is applied in order to learn a more compact and dense representation of the social group appeal. To ensure comparability across embeddings and eliminate differences driven by absolute vector magnitudes, I normalize the projected vectors by dividing them by their Euclidean norm.

$$\mathbf{v}_{sg} = \frac{\mathbf{W} \text{BERT}(P(sg))_{[\text{MASK}]} + \mathbf{b}}{\|\mathbf{W} \text{BERT}(P(sg))_{[\text{MASK}]} + \mathbf{b}\|_2} \quad (4.3)$$

All parameters of the BERT encoder as well as the weights $\mathbf{W}$ and biases $\mathbf{b}$ of the final linear projection layer are fine-tuned using the triplet loss objective originally proposed by Schroff et al. (2015). The goal of applying this objective function is to enhance the separability of social group categories in the embedding space and consequently to facilitate cleaner clustering results. For an anchor mention sg^a , a positive mention sg^+ from the same category, and a negative mention sg^- from a different category, the loss enforces that the anchor embedding is closer to the positive than to the negative by at least a margin m . Distances between embeddings are computed by applying the Euclidean distance, as defined in Equation 2.2.

$$\mathcal{L}_{\text{triplet}} = \max \left(0, d(\mathbf{v}_{sg^a}, \mathbf{v}_{sg^+}) - d(\mathbf{v}_{sg^a}, \mathbf{v}_{sg^-}) + m \right) \quad (4.4)$$

As training data for the supervised contrastive objective, I leverage the social group phrases

and categorical assignments provided within the expert dictionary (Horne et al., 2024). For each social group expression, I construct triplets by treating the phrase as the anchor sg^a and sampling another phrase from the same category as the positive example sg^+ . A negative example sg^- is then selected from a different category. I employ hard negative sampling by choosing the phrase within the batch whose current embedding representation is closest to that of the anchor term. This strategy encourages the model to learn sharper decision boundaries by explicitly separating semantically similar but categorically distinct terms. The margin parameter is set to $m = 1$, which empirically yields the most stable and robust training behavior for me. All parameters of the encoder and projection layer are fine-tuned for 30 epochs with a batch size of 32. Model weights are saved after each epoch while monitoring the triplet loss. If the loss does not decrease for 10 consecutive epochs, early stopping is applied. Ultimately, the model instance achieving the lowest training loss is selected. This training procedure is intended to increase the separability between social group categories in the embedding space and to increase the chances of recovering the social group categories of the expert dictionary. At the same time, because the model is not restricted to a fixed output space, the learned representation remains flexible enough to allow the emergence of additional group categories.

To group the social group appeals into meaningful categories, I apply the k-means clustering algorithm. This method is computationally efficient, which is particularly advantageous given the large number of parliamentary speeches analyzed. Furthermore, Genest et al. (2025) show that applying contrastive learning to entities’ embedding representations promotes the formation of roughly spherical clusters, for which k-means clustering is well suited. The algorithm requires the number of clusters k to be specified in advance. To determine an appropriate value for k , I rely on both internal and external clustering validation metrics. Internal metrics evaluate how well clusters are separated and how tightly points are grouped within each cluster, while external metrics incorporate ground-truth data for the evaluation (Rendón et al., 2011). As an internal validation metric, I use the average silhouette score, which is computed as the mean of all individual silhouette scores, defined in Equation 2.9. This metric evaluates how well clusters are separated and how tightly points are grouped within each cluster. As an external metric, I use the NMI-score, defined in Equation 2.11, which measures how well the cluster assignments reproduce known group assignments. For the NMI computation, the ground-truth data consists of all detected group mentions that can be identified using the expert dictionary through a regular expression lookup. The category assignments for each phrase then serve as input for the calculation of the score. I compute the average silhouette and NMI scores for potential k values ranging from 1 to 100. To select the optimal number of clusters, I first consider candidate values whose silhouette scores lie above the 95th percentile, prioritizing the internal metric because it has been shown to yield more accurate group detection (Rendón et al., 2011). Within this subset of candidate k values, I then choose the value that produces the highest NMI-score, incorporating external validation to ensure alignment with the known social group categories.

For each cluster, I assign a social group category term. To gain an exploratory view of the clusters’ composition, I compute the Euclidean distance from each embedding to its cluster centroid. Because the embeddings are normalized, the Euclidean distance is directly related to the cosine similarity between the cluster centroid and each respective social group mention vector. The distance measure is bounded between 0 and 2. For each cluster, I then plot these distances as a boxplot using the Plotly package (Kruchten et al., 2026). This visualization provides an immediate sense of cluster compactness and allows for the qualitative inspection of the terms most representative of the cluster as well as its outliers. Additionally, I examine the most relevant social group terms per cluster based on their tf-idf values, treating the entire cluster as a single document. Combining these two measures, I assign a qualitative label to each cluster.

I develop two supervised classification models to detect sequences of social group mentions and the author’s associated stances. This is achieved by comparing the effectiveness of different

LLM models for each task, alongside a dictionary baseline. I then apply k-means clustering to the vector representations of the detected social group appeals to categorize them into meaningful social group categories. In the following chapter, I present the performance of all models in detail and evaluate them comparatively.

Chapter 5

Model Results

I systematically evaluate different LLM architectures with respect to their suitability for classifying social group appeals. In the following sections, I present the results of this comparison, analyze where these models still fail, and illustrate how the proposed clustering approach operates on the social group terms defined in the expert dictionary.

5.1 Evaluation of Classification Models

For training the classification models, I rely on a manually annotated dataset consisting of 5,000 individual sentences. I evaluate the effectiveness of different training set sizes by examining the performance of a RoBERTa model on the cross-validated sequeval F_1 -score as a function of the number of training sentences. The results, which are presented in Appendix C, indicate that model improvements become marginal and plateau between 4,000 and 5,000 sentences. Therefore, I refrain from annotating further sentences.

As an extension of the manually annotated dataset for social group mention classification, I examine the effect of incorporating synthetic data augmentations. Table 5.1 presents the cross-validated performance of a RoBERTa model obtained by augmenting the training data with generative paraphrases of the original sentences and a combination of generative paraphrases and synonym substitutions applied to non-mention words. For both types, I add augmentations corresponding to 25% of the training set size. The results, measured in terms of the sequeval F_1 score, indicate that data augmentation does not yield substantial performance improvements for this task. The mixed augmentation strategy produces no measurable gain, while the generative paraphrases lead to only a marginal improvement. Nonetheless, as it represents an improvement, I decide to add the generatively paraphrased augmentations to the annotated dataset.

Metric	Original Data	Generative Augmentations	Mixed Augmentations
Sequeval F_1 -Score	0.81 [0.80, 0.82]	0.82 [0.81, 0.83]	0.81 [0.80, 0.83]

Table 5.1: Performance comparison of RoBERTa-base models fine-tuned on social group mention detection based on datasets synthetically augmented with different augmentation strategies. Reported are the sequeval F_1 -scores obtained by applying 5-fold cross-validation. Values in brackets represent 95% confidence intervals calculated based on the estimated standard error.

Figure 5.1 compares the best-performing model from each model family across all evaluation metrics. For clarity, I report only the F_1 -score for each metric, as it directly reflects the balance between precision and recall. A comprehensive comparison of all models within each family as well as an overview of the tuned hyperparameters is provided in Appendix C. RoBERTa delivers the strongest performance across all evaluation metrics. On the strict sequeval metric, it achieves an F_1 score of 0.82, substantially outperforming GPT-5-nano (0.65) and the dictionary baseline

(0.40). Note that the best results for GPT-5-nano are obtained by including the maximum of six few-shot examples in the input prompt. The results for the cross-span and mention detection metrics follow a similar pattern. On the cross-span metric, both RoBERTa and the dictionary baseline perform slightly better than under seqeval. This suggests that both approaches benefit from an evaluation scheme that rewards partial span matches rather than treating them as complete errors. The dictionary baseline, in particular, gains from this setup, as its predefined social group keywords rarely match the full annotated spans which oftentimes include additional modifiers. By contrast, GPT-5-nano performs slightly worse under the cross-span metric. This is primarily due to the model predicting many spans that do not even partially overlap with the annotations, while also failing to detect a substantial number of true spans. These FPs and FNs result in zero scores for individual instances which leads to an overall lower average for the generative model. When it comes to recognizing the general presence of a social group mention, all models deliver satisfactory results. Here, it is sufficient if at least one word of the original reference gets correctly predicted. RoBERTa continues to perform best with a near-perfect mention-detection F_1 -score of 0.97. Nevertheless, both GPT-5-nano and the dictionary also capture the vast majority of general mentions of social groups.

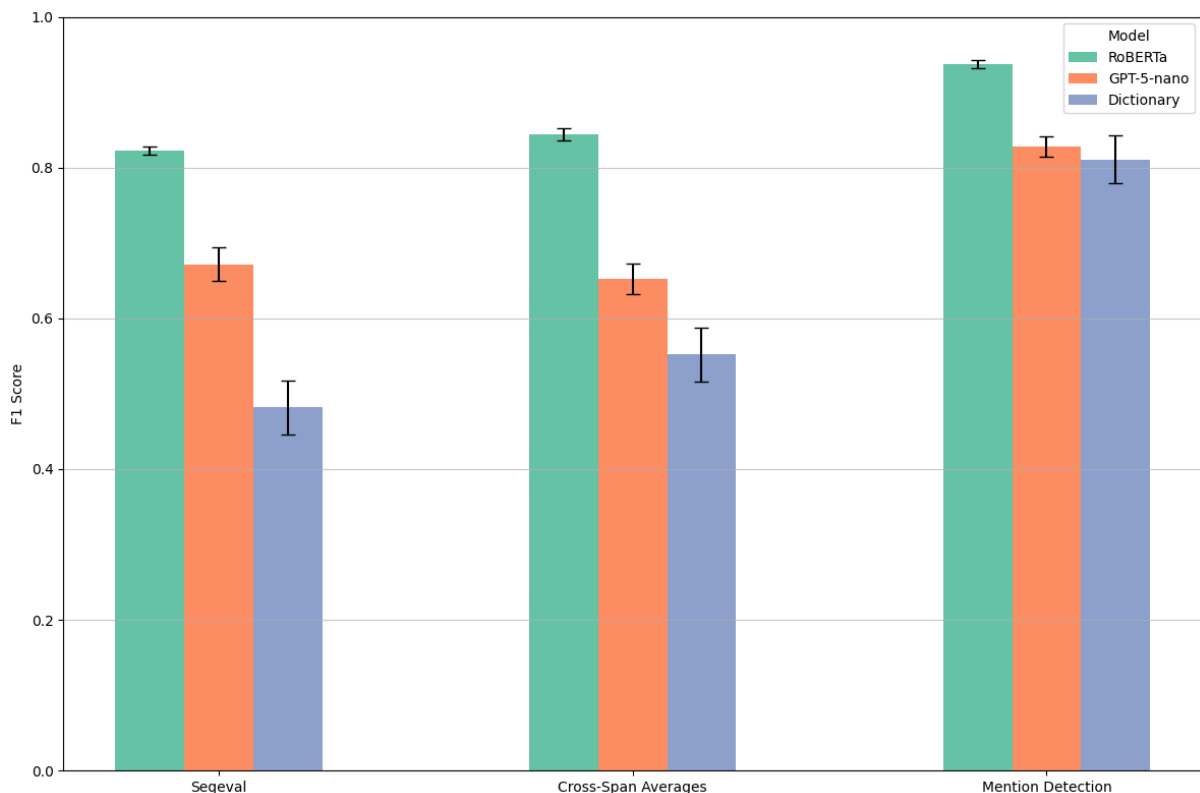


Figure 5.1: Comparison of the best performing models per model family on social group mention detection. Reported are F_1 -scores on all evaluation metrics obtained by 5-fold cross-validation. Error bars show 95% confidence intervals calculated based on the estimated standard error.

These results indicate that BERT-based models are best suited for the classification of social group appeals, with RoBERTa achieving the highest performance among the BERT variants. Table 5.2 shows a detailed summary of the cross-validated performance of the RoBERTa model across all evaluation metrics. For seqeval and the mention-detection metric, recall is higher than precision. The model detects the entire span for 84% of all social group mentions correctly and for 96% of all references at least partially. When making a prediction, the model correctly predicts the entire span in 80% of cases, while at least one word is correct in 92% of all predictions. This means that the RoBERTa model achieves to generally detect the presence of a social group

reference for the vast majority of all occurrences, but is a bit imprecise by making some false predictions. Looking at the average mention-level performance, the model correctly predicts 85% of words belonging to a social group mention, while being correct for 87% of all predicted words. Here, precision is higher than recall, which can be explained by the metric’s computation. On the mention level, the model appears to consistently predict spans that are shorter than the ground truth. When all individual words of such a prediction belong to the actual group mention, this results in perfect precision for the individual prediction but a lower recall score.

	Segeval	Cross-Span Average	Mention Detection
F1-Score	0.82 [0.82, 0.83]	0.84 [0.84, 0.85]	0.94 [0.93, 0.94]
Precision	0.80 [0.79, 0.81]	0.87 [0.86, 0.88]	0.92 [0.90, 0.94]
Recall	0.84 [0.83, 0.85]	0.85 [0.84, 0.86]	0.96 [0.95, 0.96]

Table 5.2: Summary of RoBERTa-base model performance on social group mention detection. Reported are scores on all evaluation metrics obtained by 5-fold cross-validation. Values in brackets represent 95% confidence intervals calculated based on the estimated standard error.

This hypothesis is supported by the results of a manual error analysis whose main findings are summarized in Table 5.3. I distinguish between boundary mismatches, in which predicted mention spans do not exactly align with the ground truth, false positives, where a predicted span does not correspond to any annotated span at the word level, and false negatives, where annotated spans are not recovered at all. Within each error category, I randomly sample 100 sentences to identify recurring patterns. The majority of errors fall into the category of boundary mismatches, which justifies the use of the cross-span average metric, as such partial mismatches would not be captured by segeval. Most sampled cases within this category involve predicted spans that are shorter than the ground truth, which accounts for the lower cross-span recall score. The manual inspection of the respective misclassifications shows that in most instances the model omits only an additional relative group descriptor that is not essential to the core interpretation of the social group. In only 17% of the sampled boundary mismatches does the prediction clearly exclude a substantively relevant component of the group. With respect to FPs, the model frequently predicts references that do not correspond to the conceptual definition of a social group. These include mentions of institutions such as “the police” or collections of individuals who do not share a common sociodemographic characteristic (e.g. “people who have been treated in this way”). Sometimes, the model also identifies expressions that would normally qualify as a social group mention but do not in the context of the sentence. For example, in formal greetings addressed to a colleague’s family, the model predicts the term “family” as a social group mention, even though it refers to a specific and narrowly defined collective rather than a broader social group. The analysis further indicates that indirectly formulated group references are particularly challenging for the model to detect. Such cases arise when a speaker describes a group in an implicit manner rather than referring to it explicitly (e.g. “those who carry out repair work of this kind”).

The annotated dataset does not exhibit an equal distribution of stance labels, as negative and neutral appeals are underrepresented relative to positive ones. Hence, to address this class imbalance for the training of stance classification models, I evaluate several data augmentation strategies that synthetically increase the training data of the minority classes by adding generatively paraphrased sentences. Table 5.4 summarizes the impact of these strategies on the cross-validated performance of an NLI-DeBERTa model. Only the approach of oversampling the negative class leads to an improvement over training on the original dataset alone. The overall gain is modest as the macro-average F_1 -score increases by only one percentage point compared to the baseline. Despite this limited general improvement, oversampling the negative group mentions noticeably enhances performance on that particular class. This is relevant because all

Error Category	Total Proportion	Frequent Types and Within-Proportion
Boundary Mismatch	0.55 [0.50, 0.61]	Shorter than ground truth (0.60) Longer than ground truth (0.40)
False Positive	0.25 [0.15, 0.34]	Mention of an institution (0.22) No sociodemographic feature (0.15) Specific grouping (0.11)
False Negative	0.20 [0.11, 0.28]	Indirect/descriptive mention (0.28) Mention within composite term (0.18) Abstract mention (0.12)

Table 5.3: Summary of the results of an error analysis for the social group mention detection model. Shows the total proportion of error categories as well as frequent error types based on a sample of 100 observations within each category. Values in brackets represent 95% confidence intervals calculated based on the estimated standard error.

models perform worst on the negative class, which is expected given the dataset imbalance. To enable a more balanced model performance, I adopt this augmentation strategy and oversample 25% of all negative social group appeals by adding one generatively paraphrased version per selected instance.

Metric	Original Data	Negative Class	Negative and Neutral Classes	All Classes Proportionally
Negative F1-Score	0.75 [0.67, 0.83]	0.77 [0.72, 0.81]	0.71 [0.61, 0.82]	0.73 [0.65, 0.81]
Neutral F1-Score	0.80 [0.78, 0.81]	0.79 [0.77, 0.81]	0.79 [0.76, 0.81]	0.80 [0.78, 0.82]
Positive F1-Score	0.89 [0.88, 0.90]	0.89 [0.88, 0.90]	0.88 [0.87, 0.89]	0.89 [0.88, 0.90]
Macro-Avg. F1-Score	0.81 [0.78, 0.84]	0.82 [0.80, 0.84]	0.79 [0.75, 0.84]	0.81 [0.78, 0.83]

Table 5.4: Performance comparison of NLI-DeBERTa-base models fine-tuned on stance classification based on datasets whose class imbalance got reduced by adding data augmentations adhering to different oversampling strategies. Reported are the F_1 -scores for all stance classes and their macro-average obtained by applying 5-fold cross-validation. Values in brackets represent 95% confidence intervals calculated based on the estimated standard error.

When evaluating the effectiveness of the stance classification models, it is evident that BERT-based models again achieve the best performance. Figure 5.2 illustrates that encoder-based models outperform GPT-5-nano, as the best generative LLM on the task, and the dictionary baseline across all stance categories. GPT-5-nano achieves these with the maximum number of considered few-shot examples in the prompt. The performance difference between the DeBERTa model set up as an NLI classifier and the DeBERTa model for sequence classification is relatively small, with the NLI variant performing better across all stance categories. As with the detection task, a performance comparison within model families as well as an overview about all selected hyperparameters is provided in Appendix C. The individual stance categories pose similar challenges to all model types. All models struggle most with the negative class. Here, the NLI-DeBERTa model achieves an F_1 -score of 0.76, whereas the dictionary-based approach reaches only 0.17, almost entirely failing to classify negative social group appeals. The sequence-classification BERT model performs comparably to the NLI variant, but its predictions come with much wider confidence intervals. All models perform better on neutral mentions and best on positive mentions which reflects the differences in class proportions present in the annotation dataset. GPT-5-nano is considerably better than the dictionary method for all stance labels,

but cannot compete with the BERT models. While it performs quite well for the positive class, it only achieves F_1 scores of around 0.6 for negative and neutral mentions.

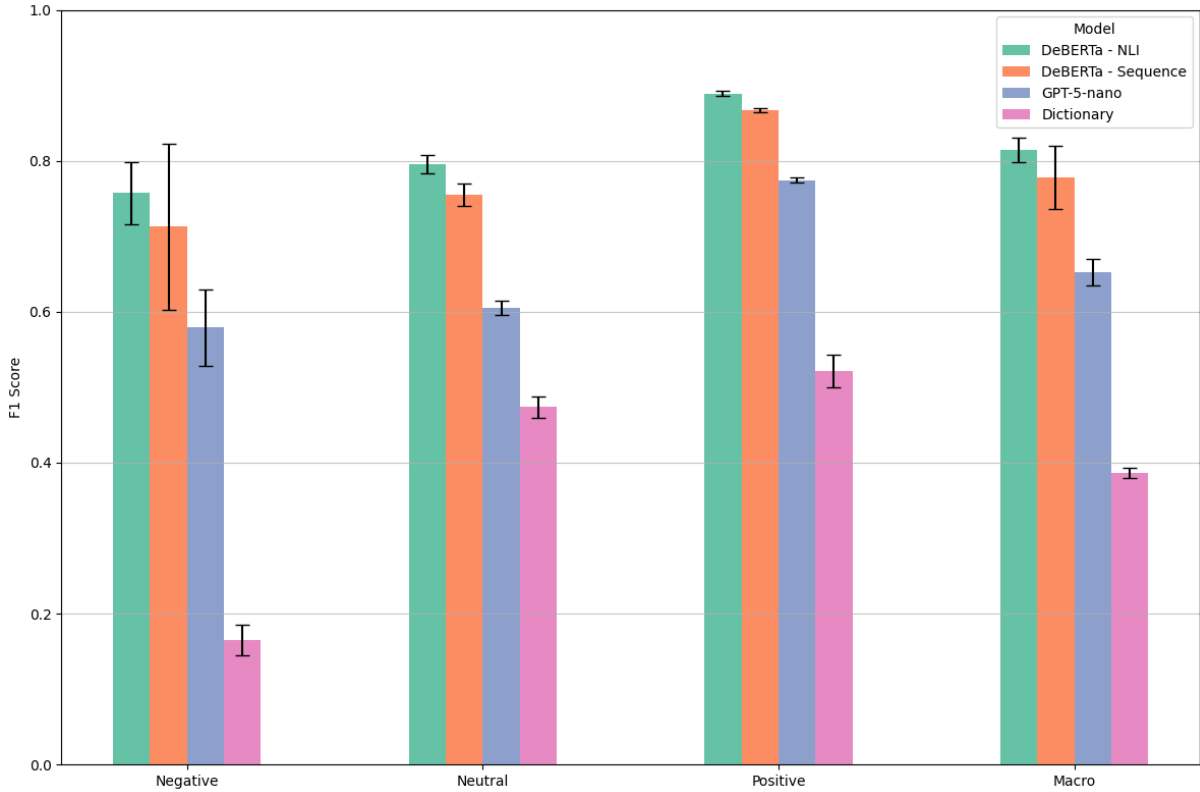


Figure 5.2: Comparison of the best performing models per model family on stance classification. Reported are F_1 -scores on all stance classes and the macro-average obtained by 5-fold cross-validation. Error bars show 95% confidence intervals calculated based on the estimated standard error.

A detailed summary of the cross-validated performance of the NLI-DeBERTa model is provided in Table 5.5. The F_1 -score for the negative class is primarily affected by limited recall. The model correctly identifies only 70% of all negative mentions. In contrast, when the model predicts a mention as negative, it is correct in 85% of cases. Both estimates are associated with relatively wide 95% confidence intervals which suggests limited certainty in the model’s performance on this class. For the neutral and positive classes, precision and recall are more balanced. 81% of all predictions labeled as neutral are correct, while the model successfully identifies 78% of all truly neutral instances. The model performs best on the positive class. 87% of positive predictions are correct, and it correctly detects 91% of all positive instances, indicating both high precision and high recall for this category.

	Negative	Neutral	Positive	Macro-Avg.
F1-Score	0.76 [0.72, 0.80]	0.80 [0.78, 0.81]	0.89 [0.89, 0.89]	0.81 [0.80, 0.83]
Precision	0.85 [0.77, 0.94]	0.81 [0.79, 0.84]	0.87 [0.86, 0.88]	0.85 [0.82, 0.87]
Recall	0.70 [0.62, 0.77]	0.78 [0.76, 0.80]	0.91 [0.89, 0.92]	0.79 [0.77, 0.82]

Table 5.5: Summary of NLI-DeBERTa-base model performance on stance classification. Reported are F_1 -scores on all stance classes and their macro-average obtained by 5-fold cross-validation. Values in brackets represent 95% confidence intervals calculated based on the estimated standard error.

To gain deeper insight into the model’s misclassifications, I present confusion matrices showing raw counts as well as row-normalized (by true labels) and column-normalized (by predicted labels) versions in Figure 5.3. The normalized matrices correspond to recall and precision scores for the respective stance classes. The results indicate that the model most frequently confuses positive and neutral appeals. Specifically, 21% of all neutral appeals are classified as positive, while 15% of cases predicted as neutral are in fact positive.

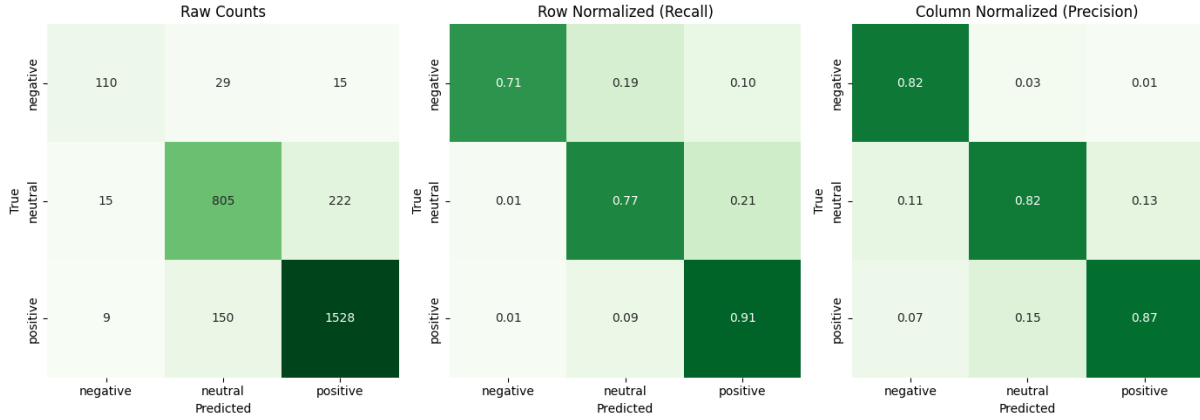


Figure 5.3: Confusion matrices for NLI-DeBERTa performance on the test sets from 5-fold cross-validation. Included are confusion matrices for the raw counts, as well as row- and column-normalized versions showing precision and recall scores.

To further investigate these error types, I randomly sample 100 instances from each of the two misclassification patterns. References predicted as positive but annotated as neutral typically occur in sentences with an overall positive tone. These sentences contain expressions of positive affect, yet the specific social group is mentioned in a neutral manner. The model nevertheless appears to rely on the broader positive sentiment and assigns a positive stance label. Conversely, positive appeals misclassified as neutral often appear in sentences framed as factual statements, for instance when the speaker cites numerical information. Although the social group is referenced in a supportive manner, the model seems to be influenced by the generally neutral tone of the sentence. These patterns underscore the challenge of distinguishing the particular stance towards an entity from a general sentence-level sentiment. Another common confusion consists of negative stances being classified as neutral. The manual inspection of this error category shows that such mispredictions often occur when negative references are expressed indirectly. In these cases, the speaker does not articulate explicit criticism but instead frames the group in a subtly negative manner. In the absence of clear negative cues, the model appears to struggle with detecting this implicit tone and therefore assigns a neutral stance label.

5.2 Evaluation of Clustering Approach

I assess the effectiveness of applying k-means clustering to the embedding space refined through contrastive learning by evaluating how accurately this approach recovers the social group categories defined in the expert dictionary. In theory, this should work exceptionally well, since the dictionary entries correspond directly to the data used for contrastive fine-tuning. The BERT model is trained to embed mentions such that those from the same category are close together, while mentions from different categories are pushed further apart. Figure 5.4 presents two-dimensional PCA projections of mention embeddings from eight example categories both before and after contrastive fine-tuning. In the original embedding space, category representations overlap substantially and occupy largely the same region. After fine-tuning, however, the groups are clearly separated. This improved structure results from minimizing the triplet loss, which

encourages embeddings of mentions within the same category to cluster tightly while enforcing a margin between them and mentions from other categories. An additional factor contributing to this separation is the use of hard negative sampling. For each triplet, the anchor embedding is paired with the most similar negative example in the batch, forcing the model to discriminate between highly confusable categories and thereby sharpening the decision boundaries in the embedding space. The figure also illustrates that the application of contrastive learning promoted the formation of spherical clusters.

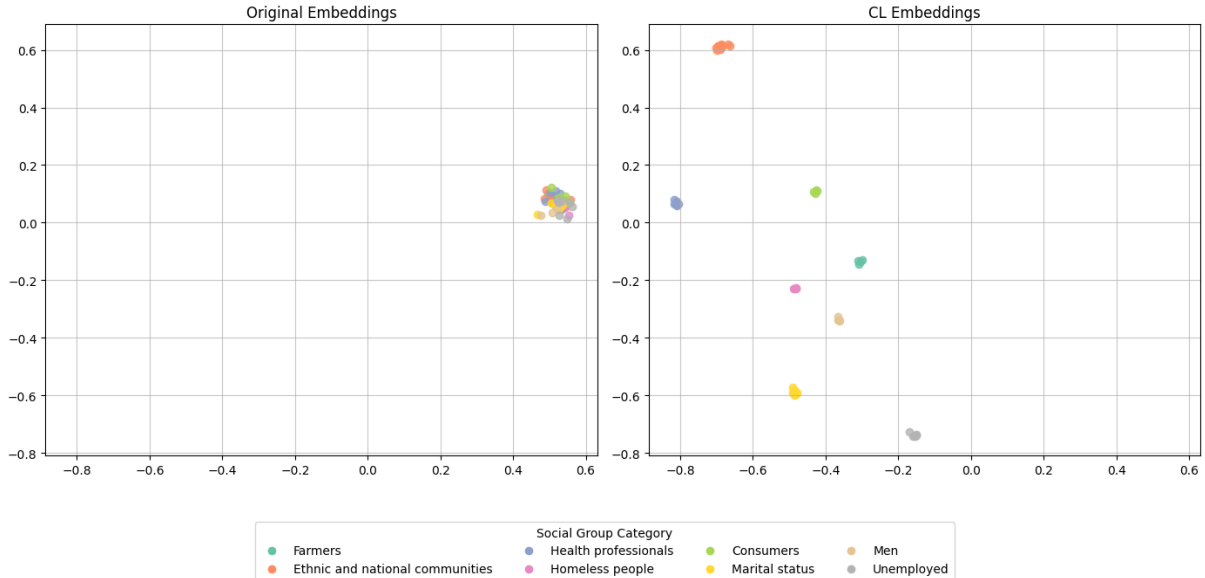


Figure 5.4: Comparison of the PCA projections of social group mention embeddings before and after the application of contrastive learning. Presented are projections for eight example social group categories highlighted by different colors.

Figure 5.5 demonstrates that applying k-means clustering to the refined embedding space for all dictionary entries allows for nearly perfect recovery of the original categories. The optimal number of clusters k is determined using both the average silhouette and the NMI score. This approach balances the objectives of producing compact, well-separated clusters while being able to accurately reproduce the ground-truth category assignments. As k increases, both metrics improve up to the original number of categories in the training data ($k = 45$). The NMI-score almost reaches the ideal value of 1.0, indicating that the resulting 45 clusters closely match the dictionary categories. Beyond this point, both metrics decline. The silhouette-score drops sharply, while the NMI-score decreases more gradually, showing that some information about category assignments is retained even with larger values of k . Overall, the trajectory of these metrics confirms that the method successfully enforces clear separation between the mention embeddings of different group categories which consequently emerge naturally as distinct clusters.

These results confirm that the method can successfully recover the social group categories defined in the expert dictionary. However, the primary goal is not only to identify these predefined categories, but also to discover additional groups that were not anticipated before. Otherwise, I could simply train a supervised classifier on the 45 predefined categories. Identifying these novel categories requires applying the method to the full research dataset. I present the results of this application in Section 7.1. As an initial check, I apply the method to all ground-truth mention spans of the manually annotated dataset. Evaluating the optimal number of clusters using the average silhouette ($\bar{s} = 0.66$) and NMI score ($\text{NMI} = 0.89$) yields a total of 49 clusters. Among these 49 clusters, several correspond to new social group categories not included in the dictionary, such as *taxpayers*, *volunteers*, or *vulnerable people*. This demonstrates that the

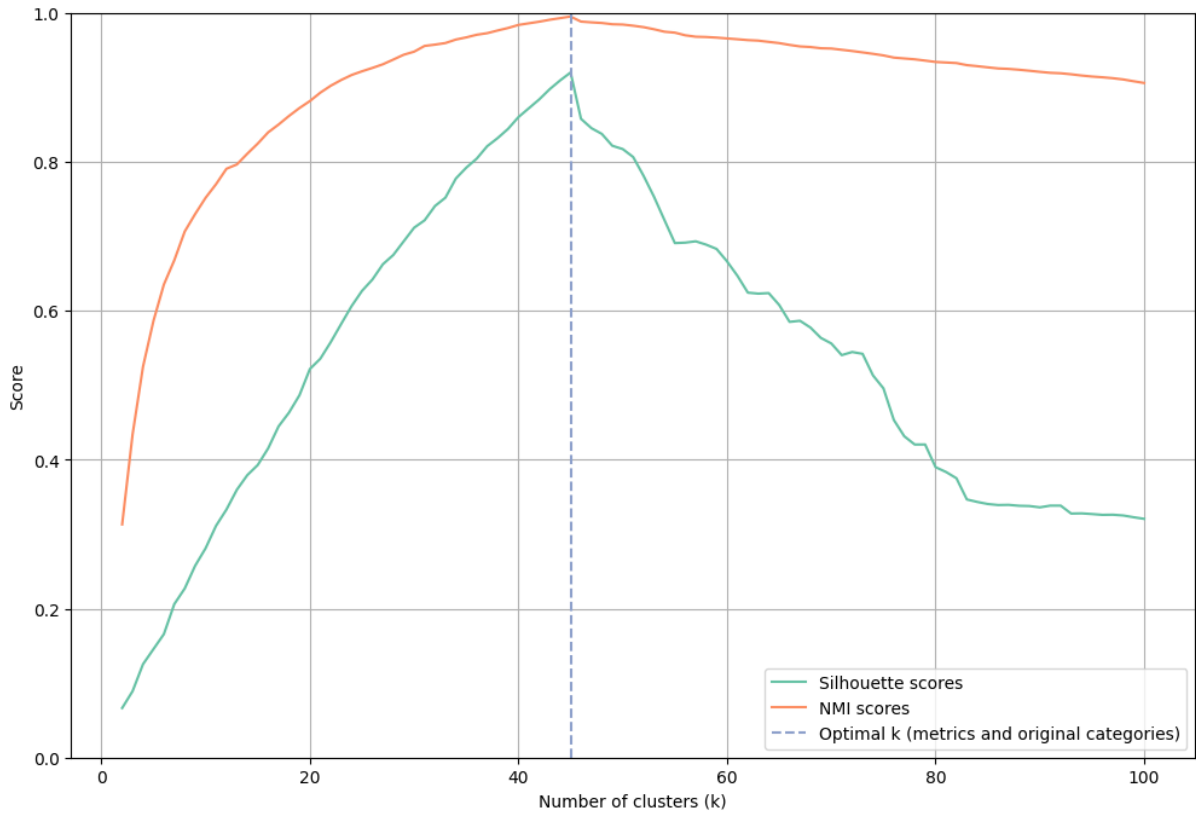


Figure 5.5: Development of average silhouette and NMI score for increasing k . Scores are obtained by applying k-means clustering on all embeddings of the social group phrases in the expert dictionary.

clustering approach can identify additional, meaningful groupings while maintaining compact and well-separated clusters.

Having evaluated the performance of the classification and clustering models for detecting social group appeals, I now outline the methodology for analyzing appeals directed at British voters' self-categorization into different age cohorts.

Chapter 6

Methodology of Empirical Analysis

I apply the developed method for detecting social group appeals to empirically examine the use and effects of age-based appeals in the UK House of Commons. Since the 2015 general election, researchers have observed a widening age divide between the Labour and Conservative parties (Curtice, 2023; Serra, 2024). While older voters predominantly support the Conservatives, Labour's support among younger voters has increased. In the following sections, I outline the hypotheses developed to examine whether this pattern is reflected in the parties' parliamentary communication and how such rhetoric affects British voters. I also describe the data sources used to test these hypotheses and formally specify the regression models employed in the analysis.

6.1 Hypotheses

Although always being relevant, age has increased in its role of a highly significant variable in explaining British peoples' vote decisions. British politics has been for decades structured around economic class as the major organizing principle for party competition and voting behavior. Caused by the strategic shift from the Labour Party during the 1990s to concentrate more on the political center and middle-class voters led to a pronounced decrease in the importance of economic status as structuring factor for British politics. This reorientation reflected itself in the parties' outward communication. The Labour Party decreased substantially its appeals to the working class (Thau, 2019) and both major parties nowadays increasingly target rather neutral, non-conflictual social groups such as families (Evans & Tilley, 2017). Due to the fact that age has taken over economic position as the biggest demographic division in UK politics, scholars recognize it as the new distinctive cleavage of British voting behavior (Curtice, 2023).

There are numerous explanations for the growing predictive power of age in explaining British election outcomes. One of them focuses on the growing ideological polarization between young and old cohorts that are taken up by Labour and the Conservatives and consequently lead to a growing gap in vote decision. Since, O'grady (2023) points out that these ideological differences between age groups are rather narrowing than widening, Serra (2024) states that it is rather the salience of ideological positions revolving around cultural and identity issues that foster their relevance in explaining voting behavior. She also points to the potential of party appeals that could play a role in the divergence.

I empirically test the role of party appeals in examining whether the age divide also reflected itself in British legislators' parliamentary rhetoric through the strategic use of positive age-based appeals. I expect that both the Labour and Conservative Party realized the growing support among the respective age groups as well as the increasing relevance of age as driver for vote decisions and consequently targeted the respective social group to a higher rate than the other party.

H1: Labour uses positive appeals targeted at young people more frequently than the Conservatives, whereas the Conservatives use positive appeals targeted at elderly people more frequently than Labour.

This hypothesis establishes expectations regarding the baseline use of age-based appeals in parliamentary rhetoric. However, MPs do not always deliver a completely unified party message since they also have further incentives to represent their local constituencies. This is especially relevant in the context of UK's electoral system whose single-member districts increase the motivations for individual legislators to take up the floor and give speeches that are also targeted at their local constituencies (Proksch & Slapin, 2012, 2015). This local focus enables them to engage in personal vote-seeking and to signal responsiveness to the interests and needs of their local voters. I expect that MPs' use of positive age-based appeals is influenced by the demographic composition of their constituencies. The share of young and elderly residents moderates the baseline differences between Labour and Conservative MPs in their use of such appeals. While I do not assume that the effect of demographic composition is limited to the party originally associated with a given age group, I expect the relationship to be stronger for that party than for the other.

H2a: A higher share of young residents in a constituency increases MPs' use of positive youth-targeted appeals, with a stronger effect among Labour MPs than among Conservative MPs.

H2b: A higher share of elderly residents in a constituency increases MPs' use of positive appeals targeted at elderly people, with a stronger effect among Conservative MPs than among Labour MPs.

With these hypotheses I test whether the age cleavage in British politics manifested itself in parliamentary rhetoric. Moreover, I analyze the effects that positive age-based appeals have on British voters. I examine whether the age-gap between Labour and Conservatives can partly be explained by the use of social group appeals targeted at the people's age. Serra (2025) shows in a survey experiment how British voters of different age groups react differently to the exposure of both symbolic and policy-related age appeals. She highlights that younger voters are more likely to support political candidates who previously spoke in favor of their age cohort. Based on the same motivation, I now apply this concept to the actual use of positive age-based appeals in parliament and the effects that these appeals have on British voters concerning their vote decision.

I expect that the age divide in voting behavior between young and old voters is reinforced by the strategic use of positive age-based appeals by MPs. When voters are exposed to appeals targeted at their own age group, these messages should strengthen existing partisan tendencies among that group relative to other voters. Consequently, increased exposure to positive youth-targeted appeals by Labour MPs should widen the gap between young and old voters in their likelihood of voting for Labour. Conversely, increased exposure to positive appeals targeted at elderly people by Conservative MPs should widen the gap between old and young voters in their probability of voting for the Conservatives.

H3a: Positive youth appeals by Labour MPs increase the probability of voting Labour among young voters more strongly than among older voters.

H3b: Positive appeals targeted at elderly people by Conservative MPs increase the probability of voting Conservative among older voters more strongly than among younger voters.

Importantly, I test these hypotheses with regard to voters' exposure to the rhetoric of their

local MP. That means that I do not link voting behavior to the collective party communication, but to the strategic communication of each MP to their constituency.

6.2 Data Sources

In order to empirically test the derived hypotheses, I draw on various data sources that provide insights into the circumstances under which legislators address social groups, as well as the effects of their rhetoric. I focus on all speeches delivered during parliamentary question debates between May 6, 2010, and December 12, 2019. I choose this time period to cover the entirety of three legislative cycles beginning with the general election 2010 and ending with the general election 2019. I focus exclusively on parliamentary question debates, as these provide especially backbenchers with an opportunity to take the floor and pose questions to the government on local issues. Furthermore, Licht and Szczepanski (2025) successfully apply the detection of social group mentions to speeches from British parliamentary question debates and show that such references are frequently used in this debate type. The text data to all speeches in the research period is provided by the ParlSpeech dataset.

I split all speeches into individual sentences and apply the developed method for detecting social group appeals to each sentence in a pipeline. First, I classify each sentence according to whether it contains a general social group mention. Next, for all detected mentions, I predict the speaker’s stance toward the referenced social group. Finally, I apply k-means clustering to the embeddings of all detected social group mentions to assign them to substantive social group categories. For this step, I use vector representations derived from a BERT-based embedding model fine-tuned via contrastive learning.

Furthermore, I supplement each speech within the research period with contextual information about the individual legislator. In addition to the complete transcripts of all speeches, the ParlSpeech dataset also contains information about the *date* on which each speech was given and the *party* to which the respective speaker belongs. In addition, I create a further variable which encodes the number of *days until the next general election* takes place. I clean the speeches’ transcripts for encoding errors and remove any superfluous leading or trailing whitespaces. In order to adequately examine the drawn hypotheses I need to rely on further context information about each individual legislator from additional data sources. To retrieve this data in a structured way, I first divide the dataset into subsets which cover a full legislative period. For each of these subsets I collect a list of all unique legislators who gave a speech in the chamber. Next, from the UK House of Commons Member API (UK Parliament, 2026) I retrieve for each legislator for each legislative period his or her *gender*, *constituency name* and information about any ministerial or opposition offices that were held within the legislative period. If no such office posts have been held, I encode the respective MP as a *backbencher*. As the API does not provide any information on the politicians’ birth date I acquire this data by requesting it from the Wikipedia API (Wikipedia, 2026). Based on the birth dates, I calculate the legislator’s *age* for each date he or she gave a speech.

Additionally, I rely on the election results for the elections 2010, 2015 and 2019 as they have been published on the UK House of Commons website (House of Commons Library, 2011, 2015, 2019). For each legislator I compute a *vulnerability* score based on the electoral advantage in percentage points ahead of the second best candidate in the previous general election. If this advantage is less than five percentage points, I encode the legislator as being electorally vulnerable. This threshold is based on Pippa Norris’ operationalization of regarding a seat as “ultra-marginal” if it was won with an advantage of less than five percentage points (Norris, 2005).

Moreover, the analysis requires information about the composition of each legislator’s local constituency. To this end, I rely on data collected in the 2011 round of the British census and published as part of the BES (Fieldhouse, Green, Evans, Mellon, & Prosser, 2022). From this

dataset, I keep all variables that provide information about the constituency’s age composition. Based on this, I compute the proportion of people *under 30* and *over 65* for each constituency.

I merge all individual data sources with the subsets containing all speakers per legislative period based on the constituency name as well as the legislator’s surname. Then, I concatenate all three subsets to gain back one dataset which covers the entire research period. There are numerous cases for which some data could not have been retrieved as the API did not provide a response or where due to some naming inconsistencies the datasets could not be merged. For all these cases I manually looked the respective information up myself and saved these corrections in additional datasets. Finally, I split each speech into individual sentences, preserving all other variables across the additional rows.

To analyze the effect of the employed group appeals on peoples’ vote decisions I rely on survey data provided by the BES. The BES conducts after each British general election a cross-sectional post-election in-person survey whose results are supposed to be representative of the general population. For this thesis, I rely on the 2010, 2015 and 2019 versions of it (Clarke et al., 2014; Fieldhouse, Green, Evans, Prosser, et al., 2022; Fieldhouse et al., 2019). From the manifold number of variables captured in the survey, I keep respondents’ *vote decision*, the *constituency name* and *code* they are located in, their *age* and *gender*. Additionally, I include the census-level variables of the *under 30* and *over 65* proportional share per constituency.

6.3 Specification of Regression Models

I test the hypotheses using two distinct regression models. To examine the use of age-based appeals in parliament, I estimate a negative binomial regression predicting the number of positive appeals directed at young and elderly people per speech. A negative binomial model is appropriate because the dependent variable is count data that takes only non-negative integer values. To account for variation in speech length, I include the logarithm of the number of sentences per speech as an offset. With this exposure adjustment, the model estimates the rate of positive age-based appeals per sentence.

$$Y_i \sim f_{\text{negBi}}(\mu_i, \theta) \quad (6.1)$$

I model the distribution of the number of positive appeals using a negative binomial rather than a Poisson distribution because the data shows overdispersion. Overdispersion implies that the Poisson assumption of the variance being equal to the expected value is violated. A likely source of this overdispersion is contagion which means that the occurrence of one group appeal increases the chances that another appeal follows. I test for overdispersion using a likelihood ratio test comparing a Poisson and a negative binomial regression model. By comparing the test statistic against the distribution of a Chi-squared distribution with one degree of freedom, I can reject the null hypothesis of equidispersion. The negative binomial model accounts for overdispersion by estimating an additional dispersion parameter θ which allows the variance to exceed the mean.

$$\text{Var}(Y_i) = \mu_i + \frac{\mu_i^2}{\theta} \quad (6.2)$$

The negative binomial regression model is a generalized linear model, meaning that the linear predictor is connected to the mean of the dependent variable through a link function. This relationship is here specified using the exponential function, which ensures that the predicted mean is always non-negative. To test the first two hypotheses, I estimate two versions of the regression equation shown in Equation 6.3. In the first model, the dependent variable is the number of positive appeals directed at young people, while in the second it is the number of positive appeals directed at elderly people. I restrict the dataset to speeches that contain either positive or negative social group appeals. Accordingly, my analysis focuses on explaining

when age-based appeals are preferred over other types of appeals, rather than on the broader conditions under which appeals occur. I exclude neutral group mentions, as these have been shown to primarily refer to factual statements without substantive attempts to appeal to the respective social group.

To test whether the effect of political party is moderated by the demographic composition of the constituency, I include an interaction between *party* and *constituency share*. Constituency share is defined as the proportion of residents under 35 living in the speaker’s constituency for the youth-appeals model and the proportion of residents over 65 for the elderly-appeals model. Since I am primarily interested in the differences between Labour and Conservative legislators I subset the dataset to contain speeches of only these two parties. Moreover, I include the speaker’s electoral *vulnerability*, *gender*, *backbencher* status as well as the *days until the next general election* as control variables because I consider them as potentially relevant for explaining the frequency of age-based appeals. The data is hierarchically clustered because the same legislators appear multiple times in the dataset. To account for this dependence, I estimate cluster-robust standard errors at the legislator level. I do not cluster at the party level because party is already included as a fixed predictor and there would be only two clusters to account for which could lead to unreliable estimations. Furthermore, I do not use a multilevel modeling approach, since the primary interest lies in the fixed effects rather than in interpreting the random structure of the data.

$$\begin{aligned} \log(\mu_i) = & \beta_0 + \beta_1 \text{party}_i + \beta_2 \text{constituency share}_i + \beta_3 \text{vulnerability}_i \\ & + \beta_4 \text{days until election}_i + \beta_5 \text{age}_i + \beta_6 \text{gender}_i + \beta_7 \text{backbencher}_i \\ & + \beta_8(\text{party}_i \times \text{constituency share}_i) + \log(\text{total sentences}_i) \end{aligned} \quad (6.3)$$

I test the effect of positive age-based appeals on British people’s vote decisions by estimating a logistic regression predicting whether a BES survey respondent votes either for the Labour or for the Conservative Party. I restrict the BES data to contain only respondents who voted for either of the two parties since my main interest lies in explaining the age differences in support for these two.

$$Y_i \sim f_{\text{Be}}(p_i) \quad (6.4)$$

I use the inverse logit transformation to transform the linear prediction to a probability between zero and one.

$$p_i = [1 + \exp(-\eta_i)]^{-1} \quad (6.5)$$

In parallel to the negative binomial models, I estimate two versions of the logistic regression model presented in Equation 6.6. In the first specification, I predict the probability that a survey respondent votes for the Labour Party. In line with *H3a*, I test whether positive youth appeals by Labour MPs increase the probability of Labour support more strongly among young voters than among older voters and thus contribute to the widening age gap in party support. I operationalize positive youth appeals as the share of all social group appeals that a Labour MP directs positively toward young people and subset the survey dataset to all respondents from Labour-held constituencies. This measure captures relative issue prioritization and directly accounts for differences in the total number of appeals made by legislators. I calculate these shares individually for all three legislative cycles and link them to the survey respondents by the election year and their local constituency.

To examine heterogeneous effects across age groups, I construct a categorical variable indicating whether the respondent is younger than 35, between 35 and 65, or older than 65. To test whether the effect of youth appeals differs across age groups, I include an interaction between the *share of youth appeals* and the respondent’s *age group*. To address potential confounding from demographic composition, where constituencies with a larger share of young residents may

both encourage youth-targeted appeals and influence the voting behavior of young people, I include the *constituency share* of residents under 35 as a control variable. In addition, I control for the respondent's *gender* and include *survey-year* fixed effects. I apply cluster-robust standard errors at the constituency level. The same general modeling strategy is applied to test the effect of appeals directed at elderly people. In this specification, the dependent variable is the probability of voting for the Conservative Party, the key explanatory variable is the share of positive appeals directed at elderly people, and the model includes the constituency share of residents over 65 as a demographic control.

$$\begin{aligned}\eta_i = & \beta_0 + \beta_1 \text{share age appeals}_i + \beta_2 \text{age group}_i + \beta_4 \text{constituency share}_i \\ & + \beta_5 \text{gender}_i + \beta_6(\text{share age appeals}_i \times \text{age group}_i) \\ & + \gamma_2 D2_i + \gamma_3 D3_i\end{aligned}\tag{6.6}$$

where,

$$\begin{aligned}D2_i &= \begin{cases} 1 & \text{if observation } i \text{ is from survey year 2017} \\ 0 & \text{otherwise} \end{cases} \\ D3_i &= \begin{cases} 1 & \text{if observation } i \text{ is from survey year 2019} \\ 0 & \text{otherwise} \end{cases}\end{aligned}\tag{6.7}$$

To facilitate the understanding of the differences between age groups beyond the changes in log-odds, I calculate the average marginal effects (AMEs) for each age group. I then calculate the differences between these AMEs and test whether these are statistically significant.

Chapter 7

Results of Empirical Analysis

I hypothesize that the age divide between young and old voters with regard to their voting decisions is reflected in parliamentary rhetoric by the major British parties through the use of strategic social group appeals. I further assume that age-based appeals exert a distinct influence on British citizens' voting behavior and thereby contribute to the widening age gap. I test these hypotheses using a series of regression models, the results of which are presented in the following sections. First, I provide a descriptive overview of the general patterns concerning the general use of appeals to social groups in the British Parliament.

7.1 Application to Speeches and Exploratory Data Analysis

I apply the best performing group mention detection (RoBERTa-base) and stance classification models (NLI-DeBERTa-base) to all speeches given during parliamentary question debates in the UK House of Commons between 2010 and 2019. The detection model finds in 89,672 out of the total of 515,768 sentences at least one social group appeal which comes down to a proportion of approximately 17%. This rather high proportion can be explained by the focus on parliamentary question debates in which legislators frequently address the concerns of diverse, oftentimes locally represented, social groups. Table 7.1 summarizes the appeal usage by the four parties with the most overall speeches. Focusing on general appeal frequencies one can observe that the Labour Party uses them most often with on-average 0.7 appeals per speech and 0.27 per sentence. The other parties all apply around 0.55 appeals per speech and 0.20 per sentence.

Party	Average Appeals		Stance Proportion		
	per Speech	per Sentence	Negative	Neutral	Positive
Conservative	0.57	0.20	0.05	0.33	0.62
Labour	0.70	0.27	0.06	0.33	0.61
Liberal Democrats	0.54	0.20	0.04	0.34	0.62
SNP	0.57	0.23	0.03	0.29	0.68

Table 7.1: Descriptive summary of appeal usage by party. Statistics are calculated for positive and negative appeals applied in parliamentary question debates within research period (2010 – 2019).

Looking at the within-party proportions of stances that are taken toward the referenced social groups, all parties are very similar to each other. They most frequently use positive social group appeals, neutral mentions in approximately one third of all cases and only very rarely negative appeals. The SNP deviates slightly from the other parties in showing with 69% an even greater share of positive appeals which leads to a slightly lower proportion of neutral mentions.

After identifying the group mention spans and predicting the stances toward them, I apply k-means clustering to the mention-level vector representations produced by the BERT-based embedding model which got fine-tuned via contrastive learning. Figure D.1 in Appendix D shows the evolution of the average silhouette and the NMI score as a function of the number of clusters k . I select the optimal value of k as the one that yields the highest NMI-score among all candidate values that fall above the 95th percentile of the average silhouette scores. This procedure results in $k = 53$, with an average silhouette score of $\bar{s} = 0.53$ and $\text{NMI} = 0.89$. These results indicate that the clustering procedure successfully recovers known social group categories and identifies additional groups, while maintaining a robust cluster structure.

I manually label each of the 53 clusters by inspecting them through boxplots of the Euclidean distances between mentions and their respective cluster centroids while examining the underlying mention texts. Furthermore, I rely on the clusters’ most representative words identified using their tf-idf scores. Figure 7.1a illustrates this labeling procedure for the social group categories “young people” and “lower class”. By highlighting individual group references in text boxes, it becomes apparent that the mentions closest to the cluster centroid capture the core semantic content of the cluster in its most basic form. As the distance from the centroid increases, the group references become more detailed and heterogeneous. These more distant mentions often include descriptive formulations, frequently expressed through relative clauses. This labeling procedure allows me to identify all social group appeals directed at young and elderly people, both of which emerge as distinct clusters and can therefore be used in the subsequent empirical analysis. Figure 7.1b presents a two-dimensional PCA visualization of the vector representations of mentions belonging to these two clusters. It illustrates that both groups are clearly separated in the embedding space.

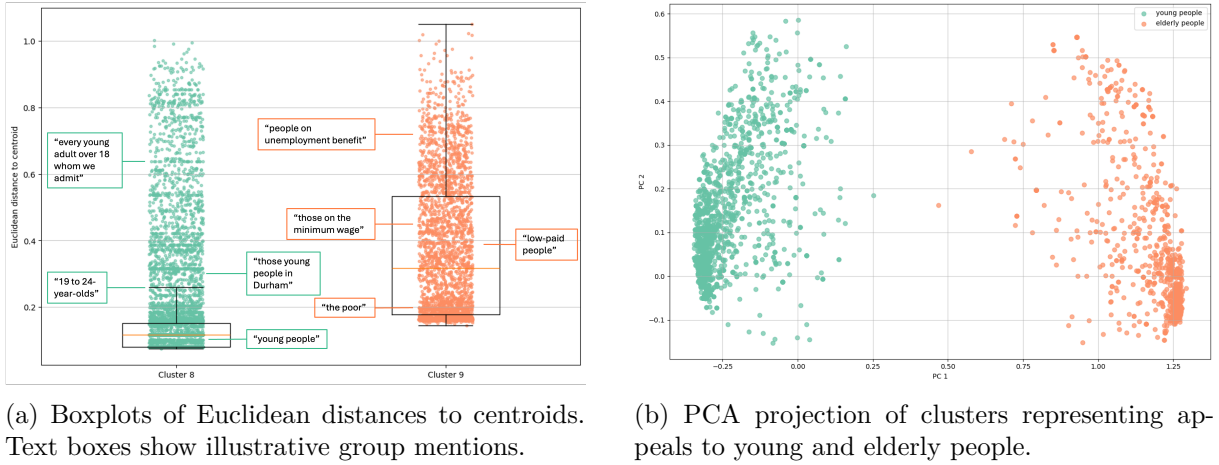


Figure 7.1: Visualizations of the social group category assignment process. (a) Boxplots of the Euclidean distances to respective centroid for all mention embeddings of two social group clusters; (b) PCA projection of clusters representing appeals to young and elderly people.

To provide more detailed insights into the social group categories that get most frequently addressed, Figure 7.2 displays the groups targeted most often by Labour and Conservative legislators, alongside the words with the highest tf-idf scores for each cluster. Several interesting observations can be drawn from this descriptive overview. Both Labour and the Conservatives most frequently address children in their communication toward social groups. This tendency is even more pronounced for Labour, which devotes approximately 8% of its total group-appeal rhetoric to this group. Labour also shows comparatively high levels of appeals to families and women. While the Conservatives also target these groups at relatively high rates, the emphasis is less pronounced than for Labour. The prominence of these groups supports the findings of Thau (2019) and Evans and Tilley (2017), who state that both parties nowadays increasingly

focus on broad, non-conflictual societal groups in order to maximize electoral support without alienating potential voters. For the Conservatives, appeals referring to criminals rank second highest, whereas such appeals are much less prominent for Labour. Interestingly, both parties show no substantial differences in their use of appeals directed at young people. This descriptive evidence against *H1* will be examined more systematically in the next section.

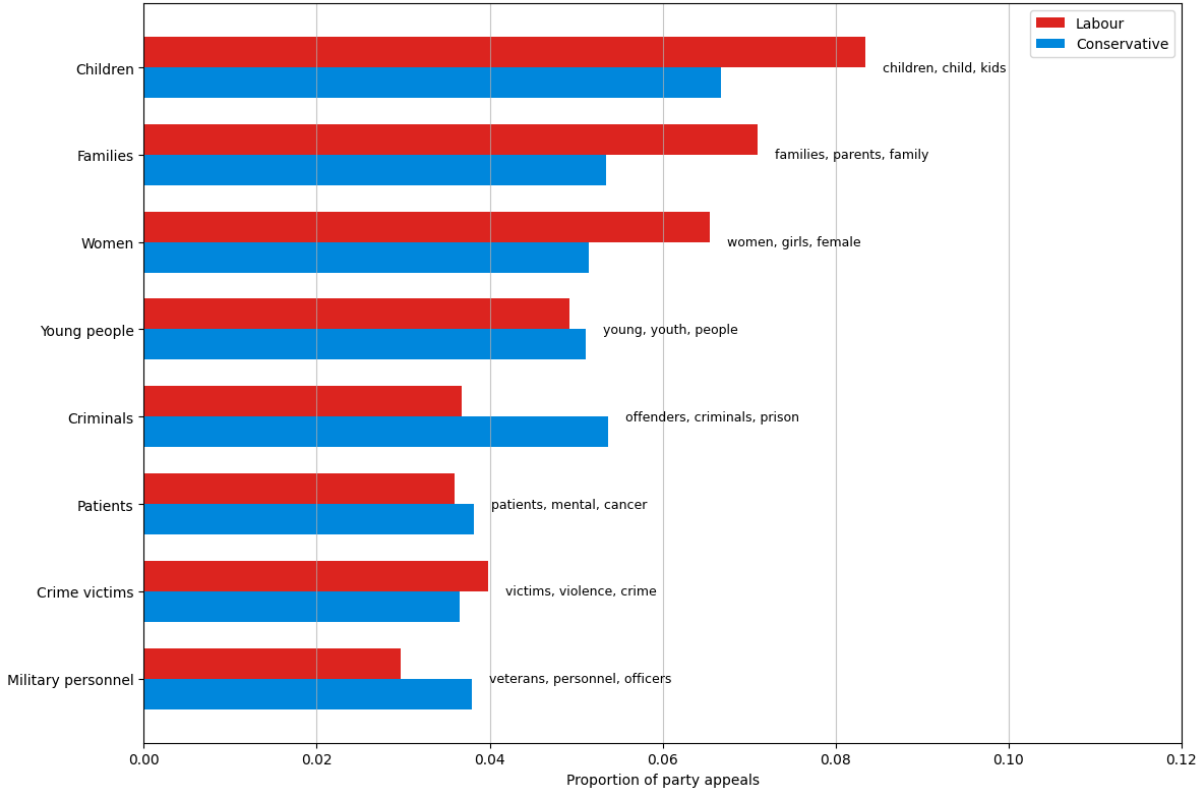


Figure 7.2: Most targeted social groups by Labour and Conservative politicians in parliamentary question debates within research period (2010 – 2019). Presented are the proportions of each social groups among all targeted groups by the party. For each group category, representative mention-words with high tf-idf scores are shown.

7.2 Use of Age-Based Appeals in the UK House of Commons

I hypothesize that the age cleavage in voter support between the Labour and Conservative parties manifests itself in parliamentary communication through the use of positive age-based group appeals. A summary of the negative binomial regressions predicting the number of youth and elderly appeals per speech is reported in Appendix D and omitted here to save some space. As a first step, I estimate models for both dependent variables without interaction terms in order to assess baseline differences between Labour and Conservative legislators in their use of age-based appeals. Contrary to expectations, no statistically significant differences are observed between the two parties in the use of appeals directed at young people. For appeals toward elderly people, however, Labour legislators are predicted to make a statistically significantly higher number of positive appeals per sentence. This finding is surprising, as the Conservatives hold higher levels of electoral support among older voters, which led to the expectation that this advantage would also be reflected in stronger targeting of elderly people in parliamentary communication. The share of young and old people is both related to a respective increase in the rate of positive appeals towards young and elderly people which partly supports *H2a* and *H2b*.

Figure 7.3 illustrates the interaction effect between party and the demographic constituency share by displaying the predicted counts of positive age-based appeals across increasing levels of demographic shares, while holding all other variables at their observed values. I employ a parametric bootstrap procedure by simulating 1000 coefficient draws from a multivariate normal distribution whose means and variances are determined by the regression model's variance-covariance matrix. These draws are then used to compute the predicted counts and their confidence intervals by calculating the bootstrap mean and the 95% quantiles for each interaction combination.

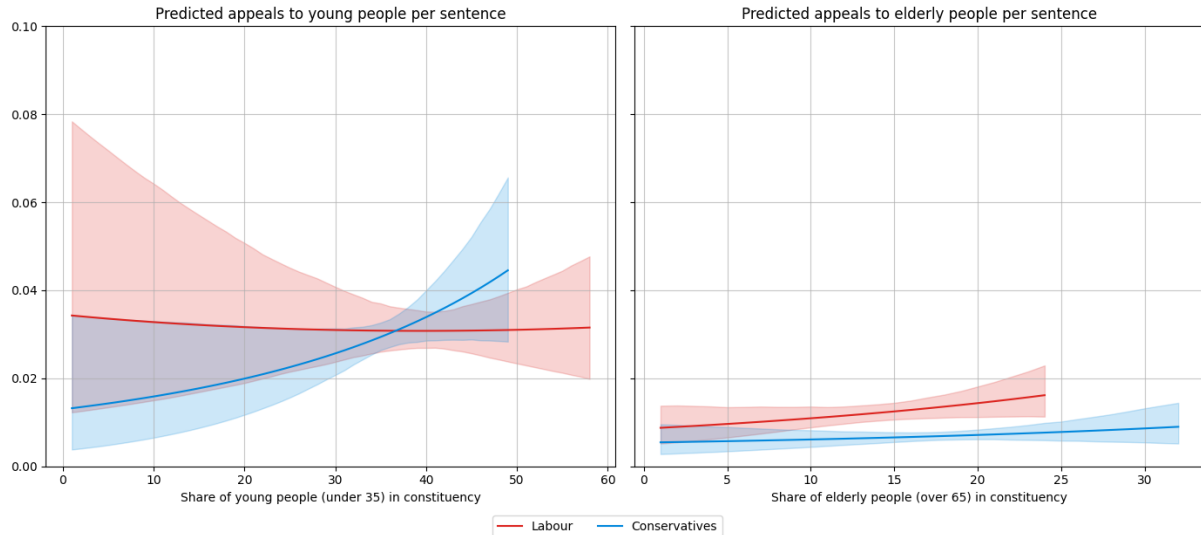


Figure 7.3: Predicted age-based appeals per sentence for Labour and Conservative MPs as a function of the demographic constituency share of young people (under 35) and elderly people (over 65). Predicted counts and 95% confidence intervals are computed via parametric bootstrapping with 1000 iterations. Confidence intervals are calculated based on the 0.025 and 0.975 quantiles of the bootstrap distribution.

Conservative legislators appear to be positively influenced by the share of young people living in their constituency when making positive appeals directed at this group. For Labour MPs, by contrast, the predicted rate of youth appeals remains largely constant across different levels of the local share of young people, albeit starting from a higher baseline level. Thus, the expectation that the local share of young people positively influences youth appeals is supported only for Conservative legislators. This finding is surprising, as I expected the demographic composition of the constituency to be particularly influential for Labour, whose traditional electorate includes younger voters.

Conversely, Conservative MPs appear to be only slightly positively influenced by the local share of people aged over 65 in their constituency. This positive relationship is more pronounced for Labour legislators, who also generally exhibit a higher baseline level. However, the difference in slopes between the two parties is small and not statistically significant. Overall, the results suggest that differences in age-based voter support between the parties are not reflected in parliamentary communication to the extent expected. While the demographic composition of constituencies appears to provide some incentive for legislators to employ age-based appeals, this relationship is modest and does not consistently point in the anticipated direction. I discuss potential explanations for these findings in the subsequent chapter.

7.3 Effects of Age-Based Appeals on British Voters

Next, I examine the effect of the strategic use of positive age-based appeals by individual legislators on the age gap in electoral support between Labour and the Conservatives. As for the negative binomial regression, the results of the logistic regression are reported in Appendix D. Figure 7.4 shows the predicted probability of voting for the Labour Party for younger and older respondents as a function of the share of positive youth appeals among all appeals made by the legislator. Because the sample includes only respondents who voted for either Labour or the Conservatives, a decrease in the predicted probability of voting for Labour corresponds directly to an increase in the probability of voting for the Conservatives. The predicted probabilities and 95% confidence intervals are computed using the parametric bootstrap method, holding all other covariates at their observed values.

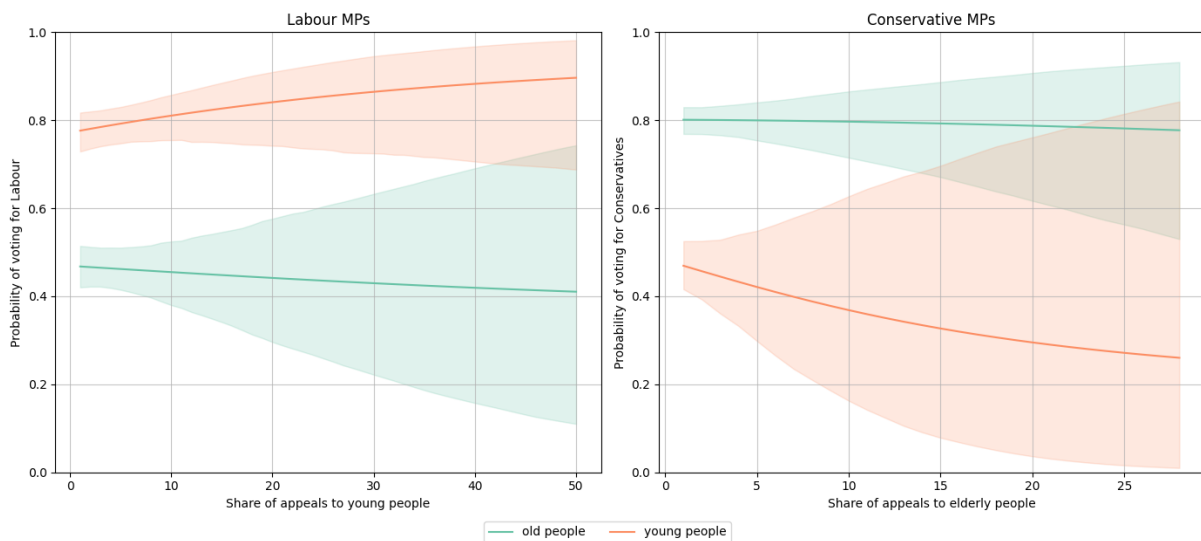


Figure 7.4: Predicted probability of vote decision as a function of the MP-level proportion of age-based appeals for different age-groups. Predicted probabilities and 95% confidence intervals are computed via parametric bootstrapping with 1000 iterations. Confidence intervals are calculated based on the 0.025 and 0.975 quantiles of the bootstrap distribution.

As expected, respondents living in a Labour-held constituency are more likely to vote for the Labour Party. The results also reveal baseline differences between young and old respondents, reflecting the described generational cleavage in party support. Higher shares of positive youth appeals are associated with an increasing probability of voting Labour among younger respondents, whereas the predicted probability remains largely constant for older respondents. This difference in slopes, corresponding to a difference in AMEs of 0.005 as shown in Table 7.2, suggests that youth-directed appeals could contribute to the age gap in support for the Labour Party. However, the magnitude of this difference is comparable to its estimated standard error and is therefore not statistically significant.

The visualization further displays the predicted probability of voting for the Conservative Party as a function of the proportion of positive appeals directed at elderly people. As in the previous analysis, the expected baseline differences in party support between younger and older respondents are visible. A stronger emphasis on appeals directed at elderly people appears to have little effect on the probability that older respondents vote for the Conservative Party. The predicted probability is already high, at around 0.8, and remains largely unchanged across different levels of elderly-directed appeals. For younger respondents, however, the slope is negative, suggesting that a stronger rhetorical focus on the elderly population may be associated with lower support for the Conservative Party among young voters. Thus, a stronger emphasis

on elderly-directed appeals by Conservative legislators appears to widen the age gap in party support, primarily because younger voters become less likely to support the party. However, this downward trend by young voters is accompanied by wide confidence intervals, indicating substantial uncertainty in the estimated effect. The difference in AMEs between old and young voters is 0.012 which highlights the widening age gap for an increase of elderly-centered rhetoric. However, as the uncertainty around the predicted probabilities already suggests, this difference is not statistically significant.

To further investigate the differences in AMEs between the two age groups, I apply parametric bootstrapping to simulate the sampling distributions of these differences. The resulting distributions are presented as histograms in Appendix D. They are approximately normal and support the just described findings.

Party	Appeal Type	Age Group Difference	AME Difference	SE	p-value
Labour	Youth appeals	young – old	0.005	0.005	0.31
Conservative	Elderly appeals	old – young	0.012	0.015	0.410

Table 7.2: Difference in average marginal effects (AMEs) between young (under 35) and old (over 65) voters for the effect of age-based appeals on vote choice. Reports the differences in the effect of youth appeals on the probability of voting Labour (young – old) as well as the differences in the effect of elderly appeals on the probability of voting Conservative (old – young).

Having presented the results of the empirical analysis, I continue by discussing the main findings and contributions of this thesis. I do so by critically evaluating the proposed method for detecting social group appeals as well as discussing the substantive findings on age-based appeals in British parliamentary communication.

Chapter 8

Discussion

After presenting the results of both the model development for measuring social group appeals and its application to age-related communication in the British Parliament, it is essential to critically discuss these findings. I do so by examining the strengths and limitations of the different models and identifying areas for improvement. In addition, I critically examine the formulated hypotheses and discuss the conclusions that can be drawn regarding the usage patterns and effects of age-based appeals by British legislators.

8.1 Critical Reflection about Social Group Appeals Measurement

The results presented in Chapter 5 clearly demonstrate that LLM-based classification models outperform dictionary baselines relying on keyword matching in detecting the span of social group mentions. This advantage holds both when evaluating strict exact-span matches as well as when allowing for partial matches. However, the dictionary approach becomes much more competitive when the task is limited to detecting the mere presence of a social group reference without considering the exact span. In this setting, it performs comparably to the GPT-5-nano model and achieves an F_1 -score of roughly 0.81. For many downstream applications in political science, detecting the general mention of a social group may already be sufficient. For example, when comparing how frequently parties generally make appeals or examining the timing or debate topic in which social groups are predominantly addressed, identifying the presence of a mention may be adequate. However, for tasks that require further classification, such as determining the stance toward the referenced group, it is beneficial to recover the full mention span so that the classification model can utilize the complete contextual information. Moreover, even at the level of simple mention detection, the best LLM model (RoBERTa) still performs considerably better than the dictionary baseline. Overall, these findings suggest that relying on an LLM-based classification model for detecting social group mentions is advisable.

When comparing generative and encoder-based LLM classification approaches for the detection of social group mentions, the results clearly indicate that encoder models are superior. All tested BERT variants outperform the best generative model (GPT-5-nano) across all applied evaluation metrics. Similar to the comparison with the dictionary baseline, these differences are most pronounced for the exact detection of the mention span. In this context, it becomes apparent that it is particularly difficult to incorporate all detailed annotation guidelines into the input prompt of the generative model. Instead, model performance is influenced more strongly by the careful selection of few-shot examples that demonstrate what qualifies as a social group and how the corresponding mention span should be extracted. Nevertheless, it appears to be both more efficient and more effective to encode this information implicitly by fine-tuning a comparatively lightweight encoder model rather than relying on extensive prompt engineering while achieving inferior performance. Consequently, it shows that is beneficial for researchers to manually an-

notate a small dataset, instead of leveraging the incorporated knowledge of a generative LLM in combination with few-shot examples. The weaker performance of generative models on discrete sequence labeling tasks is well documented in the literature and can be attributed to the fact that generative models are primarily trained to produce diverse text continuations rather than to solve discrete classification problems (Keraghel et al., 2024; Zeghidi & Moncla, 2024). At the same time, several approaches have been proposed to improve the performance of generative models on NER-like tasks (Ashok & Lipton, 2023; S. Wang et al., 2025).

Similar conclusions can be drawn for the classification of the speaker’s stance toward the referenced social group. In this case, the performance advantage of LLM-based models over dictionary approaches becomes even more evident. Capturing the stance toward a specific entity requires taking the context of the entire sentence into account. In contrast, dictionary-based methods can only rely on the overall sentiment expressed in the sentence. However, general sentiment is conceptually distinct from the stance toward a particular group. I attempted to address this limitation by restricting the analysis to a context window around the mention span, but this approach still neglects the broader contextual information contained in the full sentence. Moreover, the results indicate that leveraging a pretrained encoder model designed for NLI is beneficial compared to fine-tuning a base encoder model or using a generative LLM. The extensive training that NLI models undergo appears to provide an advantage, particularly when dealing with underrepresented stance classes such as negative social group appeals.

Adding data augmentations to the training sets for both classification tasks resulted in only minor performance improvements. Augmenting the data with sentences paraphrased by a generative LLM, however, led to some improvement in the stance classification task, particularly for the underrepresented negative class. Nevertheless, Figure C.1 amplifies that, especially for the detection of social group mentions, model performance reaches a plateau already with the original dataset of 5,000 manually annotated sentences. Adding additional original sentences appears to yield only marginal improvements. Consequently, it is not surprising that augmenting the dataset with synthetically generated variants of already existing sentences also produces only limited performance gains.

Applying clustering to categorize social group appeals into qualitative group categories offers distinct advantages, but also involves some limitations. First, embedding the group mentions using an encoder model fine-tuned through contrastive learning proved decisive for obtaining meaningful clustering results. Without this procedure, the appearing groups are considerably more ambiguous and do not exhibit a clear cluster structure. The application of contrastive learning was only possible because of the dictionary including social group categories provided by (Horne et al., 2024). Without such a carefully constructed ground-truth resource, achieving similarly reliable clustering results may be difficult, and a supervised classification approach for the specific group categories of interest might therefore be preferable. Nevertheless, the inductive detection of social group categories also offers substantial advantages. When working with large text corpora and an open-ended concept such as social group appeals, it is difficult to anticipate all possible group categories that may appear in the data. An unsupervised approach makes it possible to identify precisely such previously unconsidered categories. At the same time, the use of contrastive learning helps to ensure that previously known categories are also recovered in the clustering results.

Selecting the exact value of k when applying k-means clustering benefits from some human judgment. In this application, I determined k based on both the average silhouette and the NMI score in order to account for internal and external validation metrics. However, both metrics begin to plateau beyond a certain number of clusters, resulting in several plausible candidate values for k . Choosing among these candidates based solely on marginal differences in the validation metrics is therefore not particularly informative. A more useful approach would be to examine the resulting clusters directly to assess which value of k produces the most meaningful assignments. The investigation procedure of inspecting boxplots of Euclidean

distances to the cluster centroids alongside the most representative words in each cluster based on their tf-idf scores proved to be an effective method for manually evaluating cluster quality and selecting an appropriate value of k . I chose k-means clustering because the application of contrastive learning tends to produce roughly spherical clusters (as shown in Figure 5.4), for which this algorithm is well suited. However, future research could also explore the application of alternative clustering algorithms that do not require the manual specification of the number of clusters, such as DBSCAN. This method would also offer the benefit of directly classifying data points into a “noise” cluster so that misclassifications or group categories with only very few entries would be directly accounted for. Additionally, a density-based clustering method could be generally superior if the clustering structure is not as distinct and spherical as in the context of this application.

The methodological approach adopted in this study entails a number of limitations that could be addressed in future work. First, measuring social group appeals through a pipeline of multiple classification models introduces the well-known problem of error propagation. Errors produced by the group mention detection model are passed on to the stance classification model, which introduces its own source of error and thus compounds the overall uncertainty of the pipeline. Despite this drawback, I chose the pipeline architecture because it allows each task to be optimized independently and facilitates the targeted analysis of task-specific challenges. Nevertheless, an alternative approach would be to train a single sequence labeling model that jointly detects group mention spans and directly classifies the speaker’s stance toward them.

Second, the selection of the best-performing models for each task was not based on formal statistical significance testing. Instead, cross-validated performance scores and their confidence intervals were used to guide model selection. While the performance differences between encoder-only and generative LLMs are substantial and therefore unlikely to require formal significance testing, the performance differences among the individual BERT-based models are comparatively small. Future work could therefore examine more rigorously whether these differences reflect systematic performance advantages or are primarily due to random variation.

Moreover, the performance of the generative LLMs could potentially be improved through additional optimization steps which I did not test in this thesis. I relied on a fixed set of manually selected few-shot examples for tuning model performance. A potentially more effective approach would be to dynamically retrieve semantically similar examples from the training set for each test instance and include them as few-shot examples. Moreover, the generative LLMs were not fine-tuned but evaluated solely in a few-shot prompting setting. Although the observed performance gap between generative and encoder-based models suggests that fine-tuning would likely not close this gap entirely, investigating whether task-specific fine-tuning improves generative model performance would be interesting to examine in future work. I also need to emphasize that I only used rather small variants of the most recent closed-source GPT-models. It would be useful to test if larger generative LLMs offer substantial performance gains.

Finally, the models often face the challenge of identifying a single group mention span even when multiple, clearly distinguishable groups appear within the same text sequence. In particular, nested group mentions that occur within a more general group reference tend to be absorbed by the classification models. This oftentimes results in one predicted span that encompasses multiple group references. Consequently, the detected group mention may represent a mixture of both the specific and the broader group. Future research could address this limitation by explicitly modeling such nested structures. In this regard, approaches from the field of nested NER (Finkel & Manning, 2009) may provide useful methodological guidance, as they address a closely related problem.

8.2 UK Age-Cleavage through the Lens of Parliamentary Speeches

I examine the growing age cleavage between the Labour and Conservative parties by analyzing whether it is also reflected in their parliamentary communication and how such rhetoric affects their voters. Contrary to expectations, the analysis does not reveal the assumed baseline differences between the two parties in their use of age-based appeals in parliament. On average, Labour and Conservative legislators employ youth-targeted appeals at similar rates. Moreover, Labour MPs address appeals directed at the elderly population even more frequently than Conservative MPs, despite receiving substantially fewer votes from this age group. One possible explanation is that Labour legislators are aware of this electoral disadvantage and attempt to mitigate it through strategic appeals aimed at old voters. Conservative MPs on the other hand might not feel the need to convince their core electorate by applying respective appeals.

When examining the effect of the demographic composition of constituencies on the prioritization of age-based appeals, the results indicate that constituency demographics generally act as a modest motivating factor. In particular, Conservative MPs exhibit substantially higher proportions of positive youth-targeted appeals within their broader social group rhetoric when their constituencies contain larger shares of citizens under the age of 35. Although I expected an even more pronounced effect for the Labour Party, the predicted level of positive youth appeals among Labour legislators remains largely constant across the full range of constituency youth shares. One possible explanation is that, contrary to initial expectations, the local constituency composition may create stronger incentives for the party which is not typically associated with the respective demographic group. The Labour Party already places a relatively strong general emphasis on young citizens, which may reduce the need for individual legislators to adjust their messaging in response to local demographic variation. By contrast, youth-oriented politics are less characteristic for the Conservatives. As a result, Conservative legislators may have stronger incentives to target young voters when they constitute a substantial share of the local electorate. A similar logic could also help explain the positive relationship between the share of elderly residents in a constituency and the use of elderly-focused rhetoric among Labour MPs. Generally, structural variables appear to be more influential for explaining the use of age-based appeals. As the regression summary in Appendix D demonstrates, the age of the legislator is strongly associated with the type of appeals employed. Younger legislators tend to positively address younger citizens more frequently, whereas older MPs target older population groups at higher rates.

The descriptive statistics on the most frequently targeted social groups reinforce the findings of Thau (2019) and Evans and Tilley (2017). The groups most often addressed in parliamentary rhetoric are broad, non-conflictual societal collectives such as families, parents, or children. These categories have largely replaced traditional appeals to workers or members of the lower class, which historically constituted the central economic cleavage in British politics and were still prominently featured in Labour Party manifestos during the 1970s and 1980s (Thau, 2019). At the same time, young people do emerge as a relatively frequently targeted social group which indicates that age-based appeals represent a relevant dimension of parliamentary rhetoric.

The analysis of the effects of age-based rhetoric on differences in party support between young and old voters suggests that such appeals may contribute to the widening age gap between Labour and the Conservatives. Labour legislators appear to benefit from a stronger prioritization of positive youth-directed appeals among young voters. At the same time, old voters seem largely unaffected by these youth-oriented appeals. As a result, the age gap between young and old voters widens, but not at the expense of Labour. They do not appear to lose support among old citizens while gaining additional support among young voters. A different pattern emerges when examining appeals directed at elderly people by Conservative legislators. In this case, the age divide also widens as the share of age-related rhetoric increases. However, this effect appears to be driven primarily by a declining likelihood that young voters support the Conservatives, while the probability of support among old voters remains largely unchanged. This suggests

that emphasizing appeals to the elderly may be politically costly for Conservative MPs. Such rhetoric does not appear to increase support among their core electorate, who may be likely to vote Conservative regardless, but may instead reduce support among young voters.

I need to strongly emphasize that most of the reported effects are accompanied by substantial levels of uncertainty and are therefore not statistically significant in many cases. Nevertheless, I report these results because they may still provide informative indications of potential patterns. Nonetheless, they should be interpreted with a considerable degree of caution. Moreover, the findings should not be interpreted as evidence of causal effects. It is difficult to argue that all potentially relevant confounding variables affecting the frequency and impact of age-based appeals as well as the included independent variables have been fully accounted for. Particularly with regard to the survey analysis, future research could strengthen causal conclusions by employing matching methods that allow for a more direct comparison between structurally similar respondents who differ primarily in terms of their exposure to age-based appeals.

Finally, it is important to note that this study does not differentiate between symbolic and policy-based appeals. As Thau (2019) argues, symbolic appeals are often accompanied and reinforced by references to specific policy issues. For this reason, the analysis does not explicitly distinguish between symbolic and policy-based appeals but instead focuses on social group appeals as a broader concept that may either be purely symbolic or linked to a specific policy issue. Future research could extend this approach by developing a supervised classification method to detect whether a policy issue is mentioned alongside the group reference. A comparable approach has already been proposed by Horne et al. (2024).

The concept of social group appeals also encompasses negative rhetoric, through which political actors signal responsiveness to audiences that hold negative attitudes toward the referenced group. Although I measure negative appeals in this study, they are not examined in greater detail. Future research could therefore investigate the strategic use of such rhetoric. For instance, it would be interesting to analyze, similar to the approach taken by Riethmüller and Franzmann (2025) for social media messages, whether party status as either a dominant or a challenger party helps explain the use of negative rhetoric. Additionally, exploring party differences in the use of appeals toward criminals could provide further insights as the Conservative Party refers to this group substantially more often than Labour. Since these references are almost exclusively negative, it would be a particularly useful case for examining party differences in the prioritization of negatively framed social group appeals.

Chapter 9

Conclusion

Strategic appeals to social groups are a cornerstone of political parties’ rhetoric in their efforts to win votes (Thau, 2019). Scholars who quantify these symbolic appeals have predominantly relied on manual content analyses of comparatively small text corpora, such as party manifestos. (Dolinsky & Huber, 2023; Horn et al., 2021; Howe et al., 2022) While this approach yields valid results, the manual classification of text is highly time-intensive and does not scale well to larger volumes. In this thesis, I build on Licht and Szczepanski (2025) and Horne et al. (2024) to demonstrate the potential of LLMs for detecting social group appeals in large-scale text data. I systematically evaluate which model types are best suited for identifying the sequence span of group mentions and for classifying the author’s stance toward the referenced group. Furthermore, I apply k-means clustering to the vector representations of group mentions to derive a meaningful categorization into social group categories. These vector representations are generated using a BERT model that is further fine-tuned through contrastive learning techniques. I apply this methodological approach to analyze the use and effects of age-based appeals in parliamentary question debates in the British Parliament between 2010 and 2019. Motivated by the growing age gap in voting behavior, where young voters tend to support the Labour Party and old voters predominantly favor the Conservatives, I examine whether this divide is also reflected in parties’ parliamentary communication. Furthermore, I investigate the extent to which age-based appeals influence the gap in voting behavior between young and old voters.

The application of different LLM architectures to the classification tasks shows that encoder-based BERT models offer significant performance advantages over generative LLMs in both the detection of group mentions and the classification of the speaker’s positional stance. The findings suggest that it is difficult for the tested generative LLMs to adhere to all detailed guidelines on how to extract social group appeals that are included in their input prompt. Instead, it proved to be much more effective to manually annotate a dataset used to fine-tune a BERT model that consequently encodes all this information implicitly. Although this approach requires researchers to manually annotate individual sentences, it enables the use of comparatively lightweight open-source models that can be run locally without the need to purchase API access to closed-source generative LLMs. Dictionary approaches offer a valuable alternative to LLM-based methods when the goal is limited to identifying the general presence of a social group mention within a sentence. While still performing worse than BERT models, they nevertheless yield satisfactory results for this task. However, when it comes to detecting the exact sequence span in which the group is referenced, as well as classifying the speaker’s stance toward the group, BERT models clearly outperform dictionary approaches. In particular, stance classification requires a rich understanding of language and context, which frequency-based dictionary methods are unable to provide.

The proposed method for assigning detected social group appeals to qualitative categories using k-means clustering yields generally promising results. By refining the embedding space through contrastive learning, k-means can effectively separate group appeals into distinct clus-

ters. This is largely possible because contrastive learning tends to produce a spherical cluster structure, which k-means is well suited to capture. In addition, the combined use of boxplots displaying the Euclidean distances of group mentions to their respective cluster centroids and the inspection of clusters’ most representative terms based on their tf-idf scores enables meaningful manual labeling. However, this approach also has limitations. Contrastive learning appears to be essential for obtaining a meaningful cluster structure, as without it many group categories become conflated. This leads to the dependence on external knowledge, as provided in expert dictionaries, to guide the fine-tuning of the embedding model. Furthermore, the proposed validation metrics for selecting an appropriate number of clusters k yield a relatively wide range of plausible values. As a result, manually inspecting different candidate values appears more reasonable than relying solely on marginal differences in validation scores.

The empirical analysis of age-targeted group appeals in parliamentary speeches in the UK House of Commons between 2010 and 2019 shows that their use does not mirror the systematic differences in vote choice between different age groups. Although young voters are substantially more likely to support the Labour Party than the Conservatives, Labour legislators do not target this group more frequently than their Conservative counterparts. Moreover, contrary to expectations, Labour MPs appeal to elderly voters more often, although this group does not constitute their core electorate. This difference could be explained by the fact that Labour MPs are aware of their electoral disadvantage, which motivates them to close the gap with the Conservatives, who do not feel the need to win over their core voters even more strongly.

Interestingly, legislators from both parties appear to be more responsive to increases in the share of age groups that do not constitute a major part of their electorate. This means that Conservative MPs are more likely to positively address young people as the proportion of citizens under 35 in their constituency rises. A similar pattern is observed for Labour MPs, who increase their appeals to elderly voters as the share of citizens over 65 grows. One possible explanation for this relationship is that legislators perceive greater opportunities to gain additional electoral support among age groups that are not already part of their core voter base. A Labour MP representing a constituency with a high proportion of young voters can be relatively confident of support from this group. In contrast, the same demographic composition may provide a particular incentive for a Conservative MP to target young voters.

When examining the effect of age-based appeals on voting behavior in the UK, I find some evidence that such appeals can influence the age gap in party support. Young voters show some responsiveness to youth-directed appeals by Labour legislators, which makes them more likely to support the party as the share of these appeals increases. Old voters, on the other hand, seem to barely react to these appeals so that the age divide in voting behavior widens to some extent as a function of youth-targeted appeals. Conversely, Conservative legislators do not seem to gain comparable benefits from prioritizing appeals to elderly voters, as the older population’s likelihood of voting Conservative remains high and largely unchanged. However, young voters react negatively to these appeals and shift their support toward Labour. As a result, this communication strategy appears to be disadvantageous for Conservative MPs and further increases the voting gap between young and old citizens.

It is important to note that most of the effects described are accompanied by considerable uncertainty. Nonetheless, I consider them worth reporting, while emphasizing that their robustness should be examined more closely in future work. I further do not claim to have identified causal effects and refer to these findings rather as general relationships. Future research could benefit from applying causal inference methods, such as matching, to compare the treatment effect of different levels of age-based rhetoric among otherwise similar survey respondents. Moreover, subsequent studies should more clearly differentiate between policy-based and purely symbolic group appeals in order to disentangle the use and effects of symbolic rhetoric. The work of Horne et al. (2024) provides a useful example of how to integrate this classification into a social group appeal detection pipeline. Finally, the current empirical analysis focuses exclusively

on positive appeals in parliamentary speeches. Future research could extend this analysis by examining the determinants and effects of negative appeals. It would be interesting to investigate whether the findings of Riethmüller and Franzmann (2025), who show that challenger parties use negative group appeals more frequently than dominant parties in social media texts, also hold in the context of parliamentary speeches.

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Appendix A

Codebook

A.1 Definition of a Social Group

A social group is a segment of society or a collection of people who share common socio-demographic traits or attributes that are ascriptive and/or acquired. These include characteristics like sex and gender, age, ethnicity, language, religion, place of residence, nationality, income, occupation, education and more. Institutionally organized groups and state authorities like interest groups, trade unions, the police, and business entities are not regarded as social groups. Groupings of individuals within these institutionally organized groups are however included as long as the defining feature of the group is a common socio-demographic trait or attribute (e.g. workers, union members, police officers, business owners, teachers). Groupings based on shared beliefs, life experiences, ideology, party affiliation and/or political opinion are excluded as well as references to the people within this group are excluded.

Mentions of general groupings like “everyone” “people” or “communities” are only regarded as a social group if the term appears with a modifier that indicates a specific and defined grouping (local communities, ordinary people, our people).

A.2 Definition of a Social Group Appeal

Generally, the reference to a social group is a group mention. A group appeal is an intentional act that associates a political actor with a social group in either a supportive or critical manner. As intentional acts the interest if of explicit statements that are made in parliamentary speeches.

A.3 Stance Detection

A stance is the speaker’s position taken towards the mentioned group. The position can be either positive (favorable, supportive), negative (unfavorable, critical), or neutral.

The aim is not to code the sentiment of the entire sentence, but only the specific position the speaker expresses towards the mentioned group. Especially within questions, a stance can also be expressed indirectly or in a suggestive manner.

The sentiment annotation is done in parallel with the annotation of the group mention itself by assigning either a “sg_positive”, “sg_negative” or “sg_neutral” label.

- **Positive stance:**

The text is positive towards the group if it expresses some sort of support or positive affect. This positive position can be taken in overt as well as rather subtle ways. It can be accompanied by a concrete policy promise or just highlight the symbolic support with this group.

- **Negative stance:**

The text is negative towards the group if it expresses any form of critical or negative feeling towards the mentioned group.

- **Neutral stance:**

The text is neutral towards the group if it references the group, but neither a positive nor negative stance are taken. This can for example happen if the speaker mentions the group by reciting a neutral fact.

A.4 Clarifying Ambiguities in Classifying Social Group Mentions and Stances

Handling of Institutional Groups

- Institutional grouping based on ideology:

The definition allows for identifying groups of people within an institution but excludes those grouped primarily by ideology or party affiliation. Therefore, if a group within an institution is defined by political opinion rather than by institutional role, it should not be coded as a social group mention.

Example: I am very pleased to see the increased number of Conservative MPs who are women.

Abstract and Indirect References

- Descriptive references:

References that describe a group in abstract terms by highlighting a shared sociodemographic characteristic should be coded as a social group. These descriptions occurs most often in the form of a relative clause.

Example: The Health Secretary and I have already commissioned a review of the operation of sections 135 and 136 because we want to ensure that appropriate support and provision are available for **people who are experiencing mental health problems**, and to ensure that they are dealt with in an appropriate way.

- Abstract references:

If the reference to a group is made in an abstract form without directly naming the socio-demographic feature, this reference is regarded as a social group mention as long as it indirectly targets a socio-demographic characteristic.

Example:

- References within titles, names, programs, or institutions:

When a social group appears as part of the name of a policy program, title, statute, or piece of legislation, this indirect reference should still be coded as a social group mention. The speaker can still take a positional stance towards this group, so the stance can be either positive neutral or negative. However, since these references are most often made in factual statement the stance will be most likely neutral.

Example: The National Society for the Prevention of Cruelty to **Children** found that a third of **girls aged 13 to 17 in relationships** had experienced physical or sexual violence, with 12% of them reporting rape.

- Group references used as descriptors:

If a social group is mentioned as part of a composite term, for instance as an adjective or modifier, the reference should still be coded as a social group mention with the stance towards the group being captured.

Example: What recent estimate he has made of the effect of fiscal policy on the level of **child** poverty.

- Mentions of constituents/local residents:
Constituents or residents are not treated as implicit mentions. They are considered explicit social group mentions and should therefore be included.
Example: Too many of my **Kettering constituents** are having to wait far too long for their appeals to be heard when their benefit applications are turned down.’,

Singular versus Plural and Pronouns

- General rule:
Social group mentions generally refer to a collective of people and therefore are typically expressed in the plural form. If the speaker explicitly addresses a specific individual rather than a group, this should not be coded as a social group mention.
Example: The hon. Gentleman makes an excellent point, which I feel personally, too, as the father of an 11-year-old daughter.
- Generalization to a broader collective:
If the speaker uses the singular form to make a generalization about a broader group of people, for example through words such as “every” or “any,” this should be coded as a social group mention. Only code the group’s name and not the modifying term.
Example: I have already set out very clearly our desire to make sure that every **child** gets the best start in life.
- Singular form within a composite word:
When the singular form of a group reference appears as part of a compound or composite word, the group’s name should also be coded as a social group mention.
Example: What recent estimate he has made of the effect of fiscal policy on the level of **child** poverty.
- Pronouns:
Although a speaker may refer to a group using pronouns such as “they” “he” or “she” these mentions are not coded as social group references because the group name is not explicitly mentioned.
Example: They buy the candidates, they buy the policies and they buy the leader.

Scope of the Group

- Specific group of people mentioned: Social group mentions are directed at a larger general collective of people. References of a very specific and very small collective like the family of another legislator should not be regarded as a social group mention.
Example: I wish you and your beautiful family all the very best for the years ahead.
- Social group outside the country’s electorate:
This study focuses on social group appeals as an electoral strategy aimed at citizens of the speaker’s own country. However, references to social groups outside the domestic electorate are still coded as social group mentions because they represent clear references to identifiable groups.
Example: The revocation of article 370 of the Indian constitution without involving the **Kashmiri people** was particularly heinous.

A.5 How to Code Spans

Grammatical Nuances

- Handling of genitive forms:
When a group is mentioned using a genitive form, code the word without the genitive 's.
Example: The right to request flexible working for all and the introduction of flexible parental leave will make a real difference to **women's** lives.
- Handling of definite and indefinite articles:
Definite and indefinite articles are generally not considered part of the group mention and should be excluded when coding.
Example: I agree that that act 42 years ago was appalling, and offer my sympathies to the **families of those who were lost**.

Mention of Several Groups

- Several groups separated by conjunction:
When multiple groups are mentioned and separated by a conjunction, and the reference to each group is clear without relying on the context of the others, each group should be coded separately.
Example: Will the Prime Minister guarantee that **firefighters** and **police officers**, who we all rely on to undertake dangerous and physically demanding jobs, will retain the ability to retire and access their occupational pensions before reaching state pension age?
- Several groups not separated:
When multiple groups are mentioned together in a way that cannot be separated without losing the meaning of each individual reference, all mentions should be coded as a single social group. This often occurs when several groups are connected by a modifier that links them.
Example: 'On 11 March, I held a meeting with all the parties to discuss how best to ensure that **people with cystic fibrosis and their families** can benefit from the best drugs as soon as possible.'

Miscellaneous

- Numbers ahead of the group mention:
Numerical descriptors preceding a group reference are not considered part of the social group mention itself.
Example: In addition, he will be aware that the **Girls'** Education Challenge is helping 1.5 million **adolescent girls, who have often had children at a very young age**, to stay in education and get the education that will help to improve their lifetime earnings.
- Additional group specification in a relative clause:
When a text provides additional information about a group in a relative clause, this description should be included as part of the social group mention if it specifies the sociodemographic profile of the group.
Example: The serious question is this: how does the Secretary of State plan to monitor the impact that the removal of the bursary might have on **students from poorer backgrounds who are training as nurses in England**?

Appendix B

Model Prompts

B.1 Prompts for Social Group Mention Detection

```
System Message:
## Task Objective
Extract all qualifying social groups from the sentence as a JSON object. Do not
paraphrase the mentions at all. If no social groups are present, return an empty
JSON array.

## Definition of a Social Group
A social group is a collective of people sharing socio-demographic attributes
(age, ethnicity, income, occupation etc.).

Do not extract institutional groups, firms, state authorities, groups based on
political opinion or very specific small collectives such as one single family.
Do extract groupings of people within an institution and singular forms if it is
generalizing to a broader collective.
Extract only the social group component unless there is an additional description
to it.

User Message:
Sentence: We need to do more for young people, especially those with physical
disabilities.

Assistant Message:
{"social_groups": ["young people", "those with physical disabilities"]}

User Message:
Sentence: The Government is working in close cooperaton with the Labour Party on this
matter.

Assitant Message:
{"social_groups": []}

User Message:
Sentence: Our party supports the rights of all young people living in this country.
```

Figure B.1: Least detailed prompt for social group mention detection with two few-shot examples. Used as input for GPT-4o-mini and GPT-5-nano via OpenAI's "Responses" API.

```

System Message:
## Task Objective
Extract all qualifying social groups from the sentence as a JSON object.
Do not paraphrase the mentions at all.
If no social groups are present, return an empty JSON array.

## Definition of a Social Group
A social group is a collective of people sharing socio-demographic attributes
(age, ethnicity, income, occupation etc.).

Do not extract:
- institutional groups, political groups and state authorities
("small businesses", "armed forces", "Government", "Ministers")
- individual persons or highly specific collectives ("family of a colleague")

Do extract:
- subgroups of institutions with shared socio-demographic traits
("business people", "police officers")
- singular forms only if it is a generalization to a broader group ("every woman")
- social group component within a composite term ("victim support" -> "victim")
or title
("Society for Disabled People" -> "Disabled People")
- additional sociodemographic description of the group
("women with mental health problems")
- extract as one span if several groups are mentioned and their meaning cannot be
understood separately ("veterans, their wives and children")
- general terms ("communities", "people") only if particular group is specified
("local communities", "people with special needs")

User Message:
Sentence: We need to do more for young people, especially those with physical
disabilities.

Assistant Message:
{"social_groups": ["young people", "those with physical disabilities"]}

User Message:
Sentence: The Government is working in close cooperaton with the Labour Party on this
matter.

Assitant Message:
{"social_groups": []}

User Message:
Sentence: Our party supports the rights of all young people living in this country.

```

Figure B.2: Most detailed prompt for social group mention detection with two few-shot examples. Used as input for GPT-4o-mini and GPT-5-nano via OpenAI's "Responses" API.

B.2 Prompts for Stance Classification

```
System Message:
## Task Objective
Label the following sentence according to the stance the speaker expresses towards the
highlighted social group. Return this label as a JSON object.
The stance can be either positive, neutral or negative.

## Positive Stance
The sentence expresses support or positive affect towards the group. Especially within
questions, a positive stance can also be expressed indirectly by raising the
interests of the group.

## Neutral Stance
The sentence references the group, but neither a positive nor negative stance
are taken. This happens mostly within factual statements.

## Negative Stance
The sentence expresses a critical or negative feeling towards the mentioned group.
Raising awareness of the disadvantage faced by a group is not a negative but a positive
stance.

User Message:
Stance towards young people: Our party stands for improving the job opportunities of
young people.

Assistant Message:
{"stance": "positive"}

User Message:
Stance towards GP: Each person in this country should have the chance to get an
appointment at his or her GP within a couple of days.

Assistant Message:
{"stance": "neutral"}

User Message:
Sentence: Stance towards criminal offenders: We must do everything we can to tackle
violence on our streets and get criminal offenders into jail.

Assistant Message:
{"stance": "negative"}

User Message:
Stance towards students: We endorse the rights of the young students in this country.
```

Figure B.3: Least detailed prompt for stance classification with three few-shot examples. Used as input for GPT-4o-mini and GPT-5-nano via OpenAI's "Responses" API.

```

System Message:
## Task Objective
Label the following sentence according to the stance the speaker expresses towards the
highlighted social group. Return this label as a JSON object.
The stance can be either positive, neutral or negative.
Label the sentence according to the following definition and criteria.

## Positive Stance
The text is positive towards the group if it expresses some sort of support or positive
affect. Especially within questions, a positive stance can also be expressed indirectly
by raising the interests of the respective group or criticizing their disadvantage.

## Neutral Stance
The text is neutral towards the group if it references the group, but neither a
positive nor negative stance are taken. This happens mostly when the speaker mentions
the group by stating a fact or when the group is part of a title or organization name.

## Negative Stance
The text is negative towards the group if it expresses any form of critical or negative
feeling towards the mentioned group. Raising awareness of the disadvantage faced by a
group is not a negative but a positive stance.

User Message:
Stance towards young people: Our party stands for improving the job opportunities of
young people.

Assistant Message:
{"stance": "positive"}

User Message:
Stance towards GP: Each person in this country should have the chance to get an
appointment at his or her GP within a couple of days.

Assistant Message:
{"stance": "neutral"}

User Message:
Sentence: Stance towards criminal offenders: We must do everything we can to tackle
violence on our streets and get criminal offenders into jail.

Assistant Message:
{"stance": "negative"}

User Message:
Stance towards students: We endorse the rights of the young students in this country.

```

Figure B.4: Most detailed prompt for stance classification with three few-shot examples. Used as input for GPT-4o-mini and GPT-5-nano via OpenAI's "Responses" API.

Appendix C

Detailed Classification Results

C.1 Dataset Size Evaluation

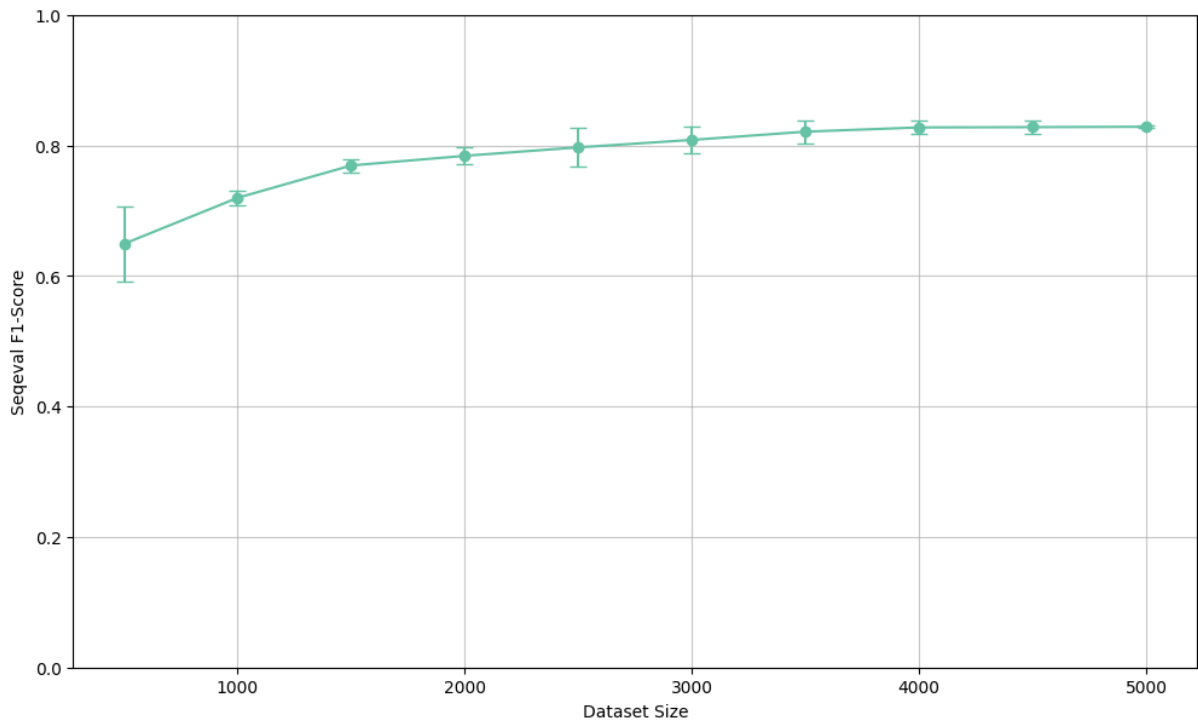


Figure C.1: Performance development of fine-tuned RoBERTa-base model on sequeval F_1 -score with increasing dataset size. Reported are average test set performances obtained by 5-fold cross-validation.

C.2 Comparison within Model Families - Social Group Mention Detection

Model	Learning Rate	Weight Decay	Batch Size	Epoch
BERT-base-cased	$2e^{-05}$	0.30	32	7
roberta-base	$2e^{-05}$	0.10	16	8
distilbert-base-cased	$9e^{-06}$	0.01	8	9
microsoft/deberta-v3-base	$4e^{-05}$	0.01	8	8

Table C.1: Overview of the tuned hyperparameters for all BERT models on social group mention detection. The hyperparameters are searched for 20 trials with the TPE-algorithm implemented within the Optuna package (Akiba et al., 2019) via application to the validation set.

Model	Temperature	Reasoning Effort
GPT-4o-mini	0	Not specified
GPT-5-nano	Not specified	High

Table C.2: Overview of the tuned hyperparameters for all GPT models on social group mention detection. The hyperparameters are greedily searched by comparing their performance on the validation set.

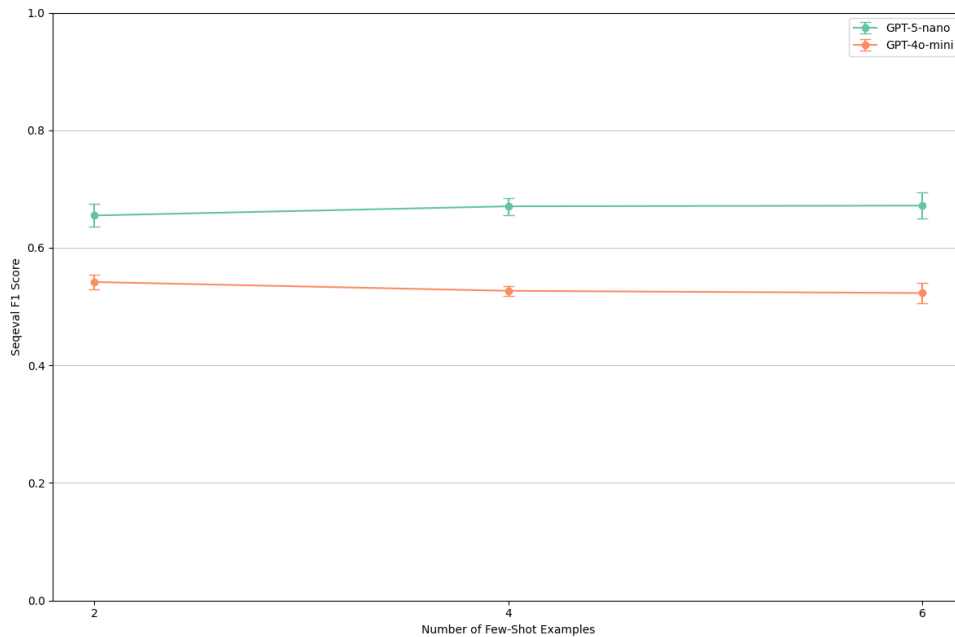


Figure C.2: Development of seqeval F_1 -score on social group mention detection over increasing number of few-shot examples. Performances are presented for GPT-4o-mini and GPT-5-nano. Error bars represent 95% confidence intervals that are calculated based on the estimated standard error.

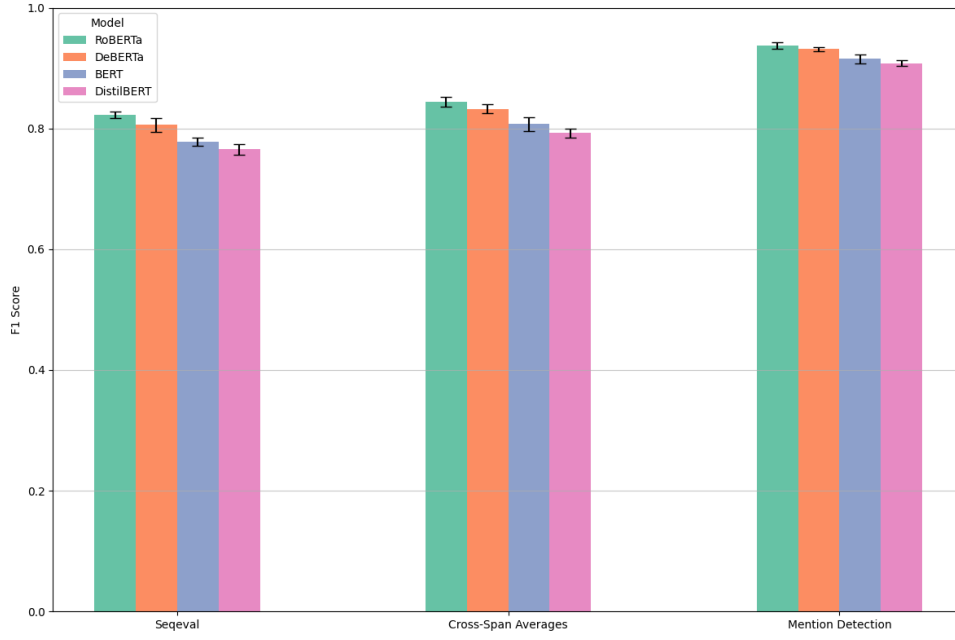


Figure C.3: Comparison of all BERT-based social group mention detection models. Reported are F_1 -scores on all evaluation metrics obtained by 5-fold cross-validation. Error bars represent 95% confidence intervals that are calculated based on the estimated standard error.

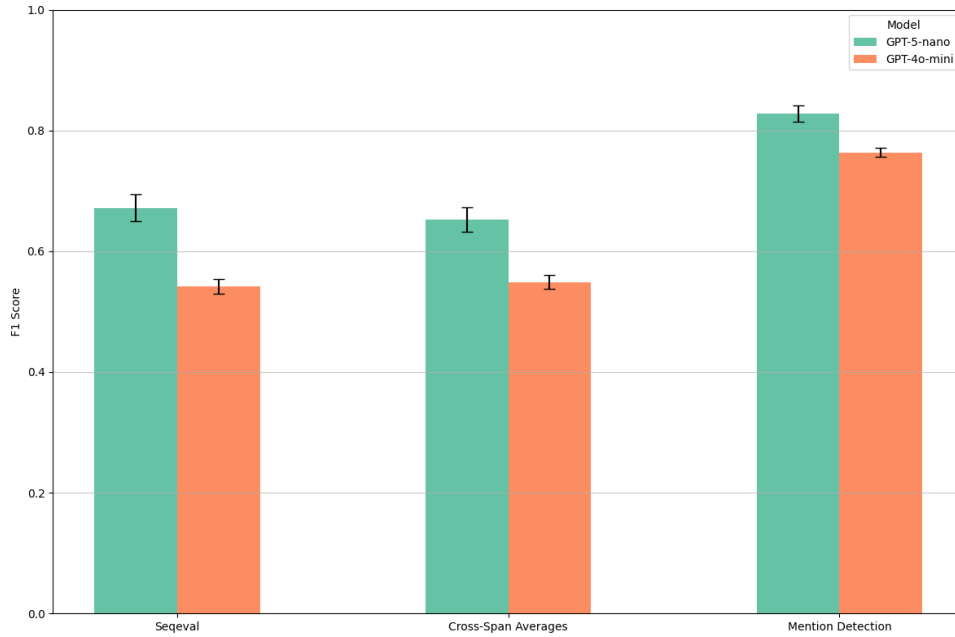


Figure C.4: Performance comparison of all generative LLMs for social group mention detection. Reported are F_1 -scores on all evaluation metrics obtained by 5-fold cross-validation. Error bars represent 95% confidence intervals that are calculated based on the estimated standard error.

Model	Learning Rate	Weight Decay	Batch Size	Epoch
BERT-base-cased	$2e^{-05}$	0.01	9	9
roberta-base	$4e^{-05}$	0.00	16	8
distilbert-base-cased	$2e^{-05}$	0.3	32	8
microsoft/deberta-v3-base	$9e^{-06}$	0.1	8	6
NLI - microsoft/deberta-v3-base	$2e^{-05}$	0.3	16	4

Table C.3: Overview of the tuned hyperparameters for all BERT models on stance classification. The hyperparameters are searched for 20 trials with the TPE-algorithm implemented within the Optuna package (Akiba et al., 2019) via application to the validation set.

Model	Temperature	Reasoning Effort
GPT-4o-mini	0.5	Not specified
GPT-5-nano	Not specified	High

Table C.4: Overview of the tuned hyperparameters for all GPT models on stance classification. The hyperparameters are greedily searched by comparing their performance on the validation set.

Dictionary	Window Size	Positive Threshold	Negative Threshold
Lexicoder Sentiment Dictionary	6	0.4	-0.4

Table C.5: Overview of the tuned hyperparameters for the dictionary approach on stance classification. The hyperparameters are greedily searched by comparing their performance on the validation set.

C.3 Comparison within Model Families - Stance Classification

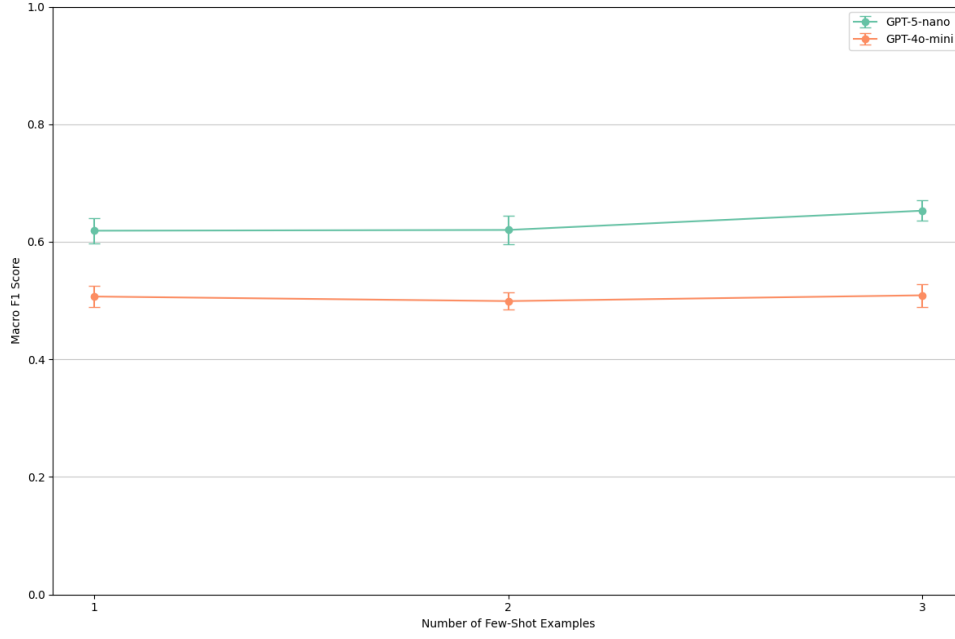


Figure C.5: Development of macro-average F_1 -score on stance classification over increasing number of few-shot examples. Performances are presented for GPT-4o-mini and GPT-5-nano. Error bars represent 95% confidence intervals that are calculated based on the estimated standard error.

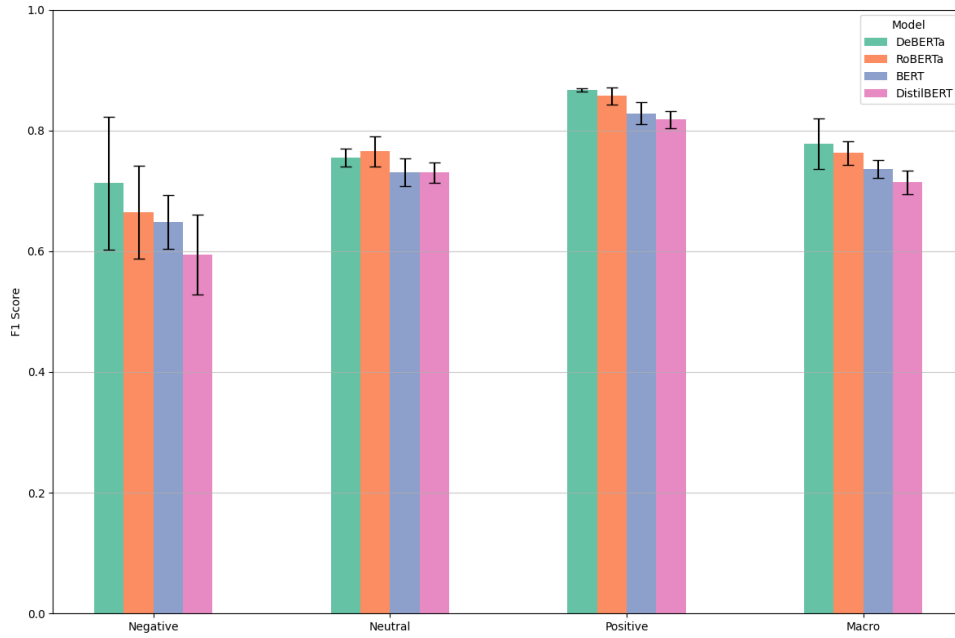


Figure C.6: Comparison of all BERT-based sequence models for stance classification. Reported are F_1 -scores on all stance classes and the macro-average score obtained by 5-fold cross-validation. Error bars represent 95% confidence intervals that are calculated based on the estimated standard error.

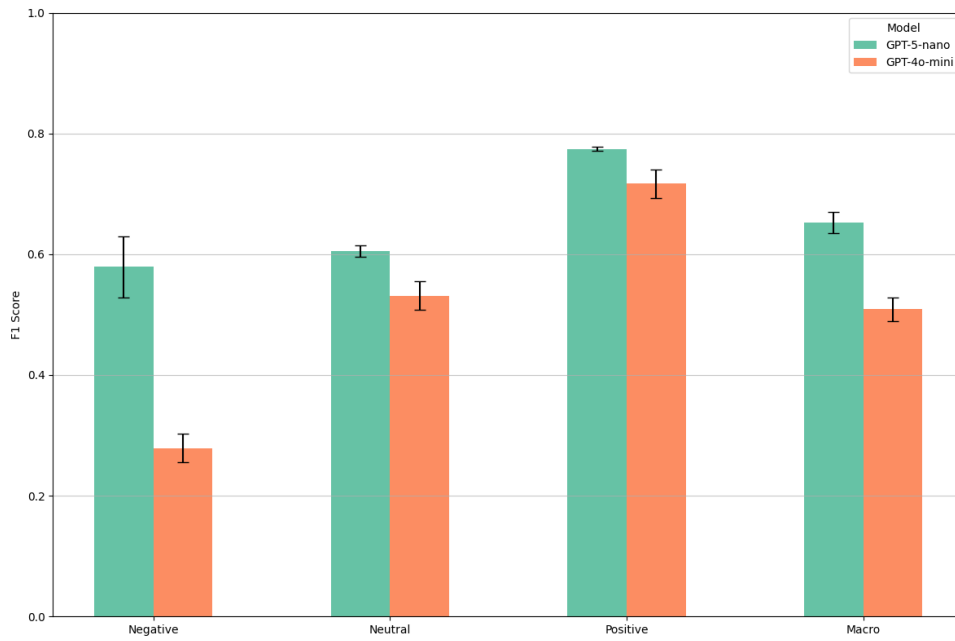


Figure C.7: Performance comparison of all generative LLMs for stance classification. Reported are F_1 -scores on all stance classes and the macro-average score obtained by 5-fold cross-validation. Error bars represent 95% confidence intervals that are calculated based on the estimated standard error.

Appendix D

Detailed Empirical Results

D.1 Clustering Results

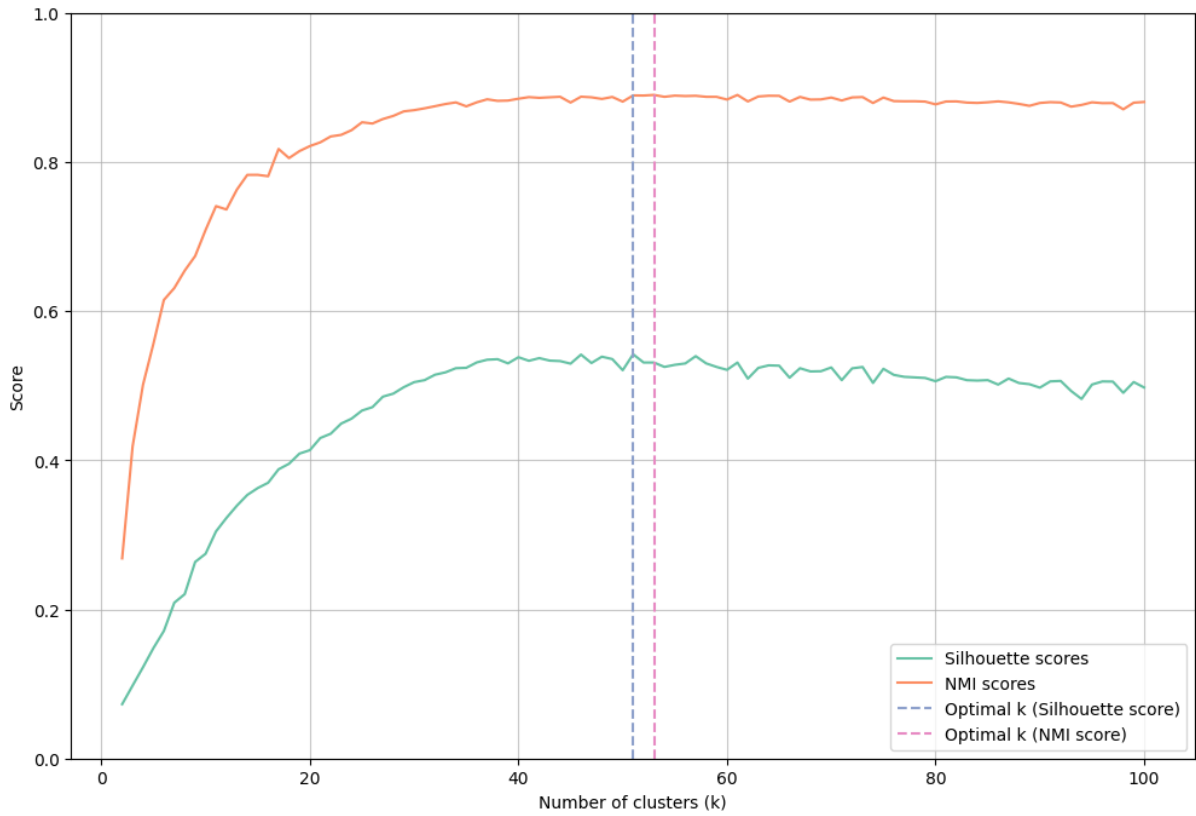


Figure D.1: Development of average silhouette and NMI score for increasing k . Scores are obtained by applying k-means clustering on all embeddings of the social group phrases detected in parliamentary question debates within research period (2010 – 2019).

D.2 Regression Results

	Appeals to Young People		Appeals to Elderly People	
	(1)	(2)	(1)	(2)
Intercept	-3.285*** (0.413)	-3.828*** (0.650)	-6.160*** (0.371)	-6.070*** (0.418)
Labour Party	-0.030 (0.101)	1.008 (0.780)	0.656*** (0.115)	0.466 (0.435)
Constituency Share	0.013 (0.010)	0.028 [†] (0.017)	0.021 [†] (0.013)	0.016 (0.019)
Vulnerable	0.077 (0.102)	0.049 (0.101)	-0.163 (0.144)	-0.172 (0.142)
Days until election	0.00007 (0.00009)	0.00007 (0.00009)	0.00015 (0.00012)	0.00015 (0.00012)
Age	-0.018*** (0.004)	-0.018*** (0.004)	0.010* (0.005)	0.010* (0.005)
Female	0.080 (0.089)	0.072 (0.087)	-0.360*** (0.103)	-0.361*** (0.103)
Backbencher	0.246** (0.082)	0.245** (0.082)	0.522*** (0.117)	0.521*** (0.117)
Labour × Constituency Share		-0.028 (0.021)		0.012 (0.025)
Theta	0.128 (0.005)	0.129 (0.005)	0.047 (0.003)	0.057 (0.003)
Observations	60849	60849	60849	60849
Log-Likelihood	-17170.15	-17165.9	-6383.16	-6382.999
AIC	34358.3	34351.8	12784.32	12786

Note: Constituency share is the share of people < 35 for appeals to young people and the share of people > 65 for appeals to elderly people. Reference categories are the Conservative party, non-vulnerable legislators, male legislators and frontbenchers. Cluster robust standard errors at the MP level are presented in parentheses.

[†] $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table D.1: Negative binomial regression results predicting the number of appeals targeted at the youth and elderly people per sentence. Application of cluster robust standard errors on the level of individual MPs.

	Pr(Vote Labour)	Pr(Vote Conservative)
Intercept	0.533 (0.470)	-0.278 (0.334)
Share age appeals	0.0240 (0.019)	-0.052 (0.060)
Middle-aged voter	-0.717*** (0.154)	0.775*** (0.120)
Old voter	-1.375*** (0.154)	1.499*** (0.135)
Constituency share	0.023* (0.011)	0.023 (0.016)
Female voter	0.182 [†] (0.103)	-0.194* (0.090)
Election year: 2017	0.056 (0.136)	-0.475*** (0.110)
Election year: 2019	-0.508*** (0.130)	0.066 (0.122)
Share age appeals × Middle-aged voter	-0.024 (0.019)	0.050 (0.057)
Share age appeals × Old voter	-0.030 (0.023)	0.050 (0.064)
Observations	4488	4488
Log-Likelihood	-2679.807	-2677.049
AIC	5391.613	5386.098
Note: Share age appeals is the share of appeals targeted at young voters for the first (Pr(Vote Labour)) and the share of appeals targeted at elderly voters for the second (Pr(Vote Conservative)) regression. Constituency share is the share of people < 35 for the first and the share of people > 65 for the second regression. Reference categories are young voters, male voters and the 2015 election year. Cluster robust standard errors at the constituency level are presented in parentheses. [†] $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$		

Table D.2: Logistic regression results for predicting the effects of age-based appeals on the probability of voting Labour and Conservative. Coefficients are reported in log-odds. Share age appeals denotes the share of youth/elderly appeals applied by the MP elected for the respective voter's constituency in parliament.

D.3 AME Bootstrap Distributions

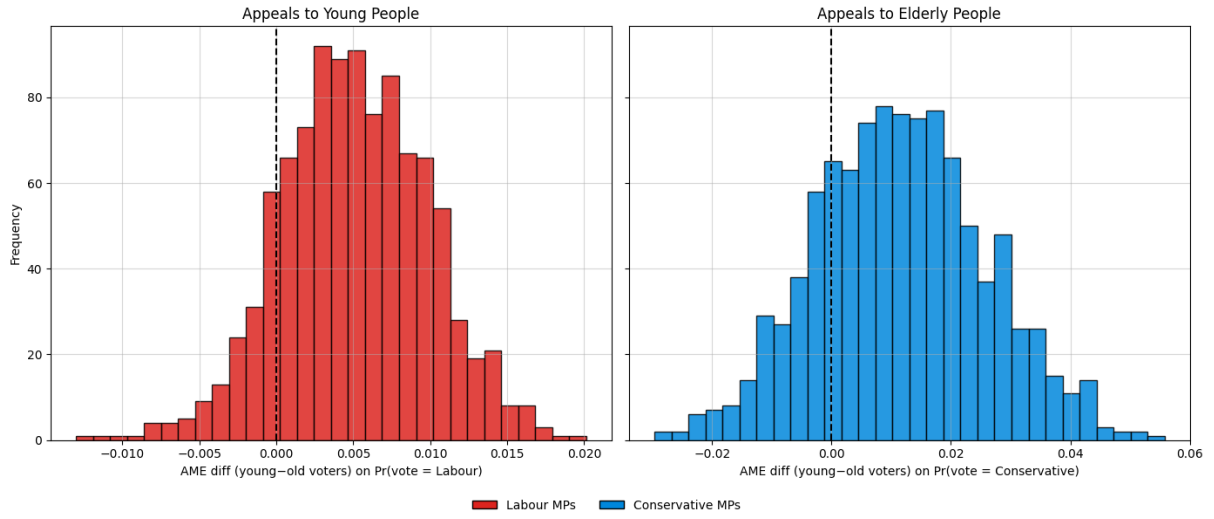


Figure D.2: Bootstrap distributions for the difference in average marginal effects (AMEs) between young (under 35) and old (over 65) voters for the effect of age-based appeals on vote choice. Reports the differences in the effect of youth appeals on the probability of voting Labour (young – old) as well as the differences in the effect of elderly appeals on the probability of voting Conservative (old – young). Parametric bootstrapping with 1000 iterations is applied.